BUSINESS COMMUNICATION

Coursepack

Name:

ID:

Section:

Department:

Semester:

Mid Term

Writing E-mails, Letters, and Memos

Types of E-mails, Letters, and Memos

6.1 Recognize the basic types of correspondence.

All three types of correspondence are used for similar purposes, so their content, organization, and style are also similar. Here is how they differ:

E-mail is the workhorse of the scientific and technical workplace. It is used to ask questions, provide information, and explain decisions.

Letters are written to people *outside* the company or organization. Primarily, letters are used in formal situations in which an employee is acting as a representative of the company. Letters can be used to make requests or inquiries, accept or refuse claims, communicate important information, record agreements, and apply for jobs.

Memos are written to people *inside* the company or organization. They are used to convey decisions, meeting agendas, policies, internal reports, and short proposals. When a message is too important or proprietary for e-mail, most people will send a memo instead. Memos are still more reliable than e-mails for information that should not be broadly released.

You will occasionally hear that texting and social networking are making e-mails, letters, and memos obsolete. That's really not true. At almost all companies, e-mail is still the primary way information is shared and decisions are recorded.

Meanwhile, letters and memos are transforming into electronic documents (e.g., PDF files) that can be viewed on screens. The electronic format allows important decisions to be formalized and archived because PDFs are difficult to alter and they can be electronically signed and password protected.

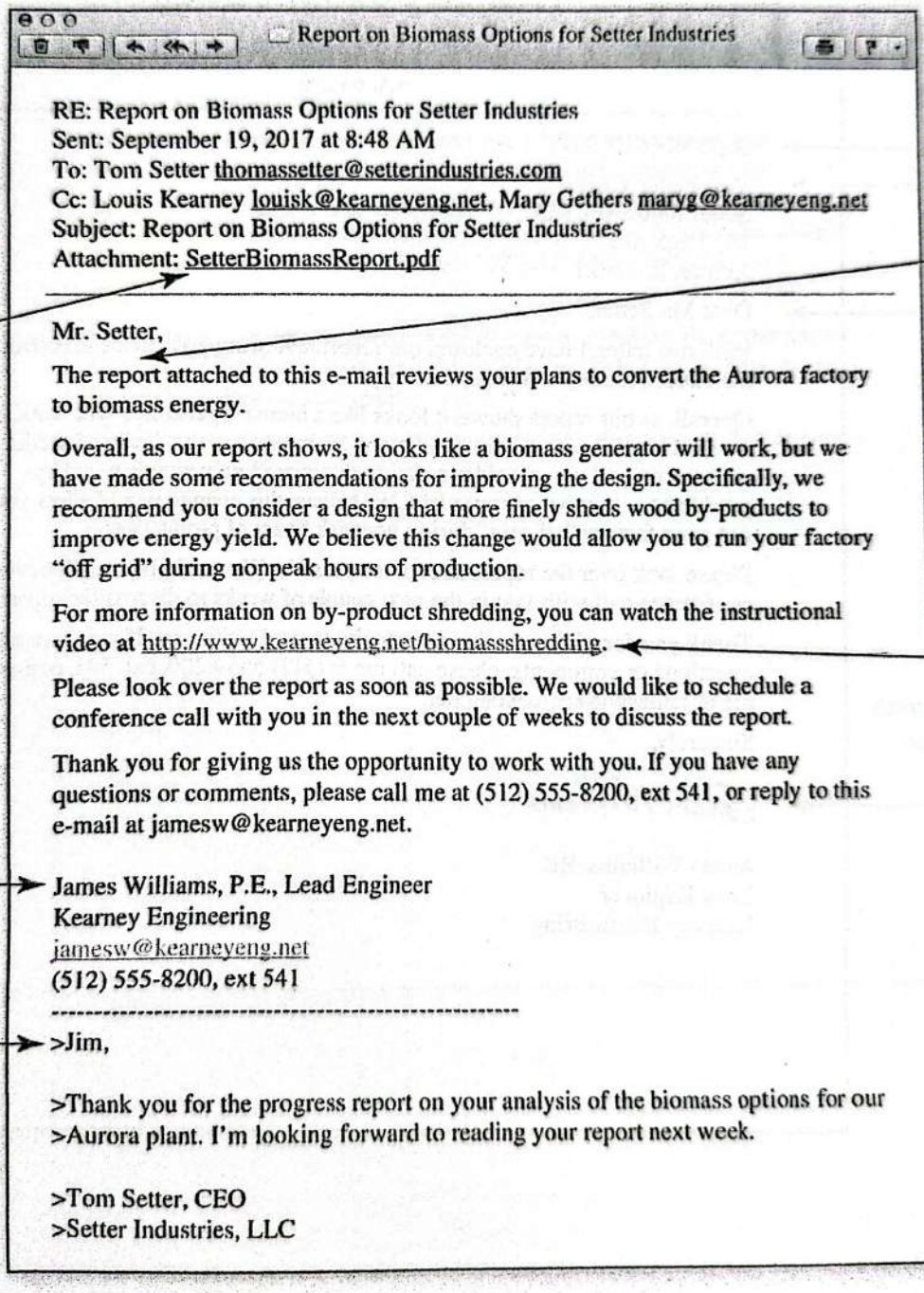
E-mails, letters, and memos primarily differ in the way they are *formatted* (how they appear on the page or screen). Figure 6.1 shows how the same information would look in e-mail format, letter format, and memo format.

- An e-mail includes a header with an addressee line for the readers' e-mail addresses ("To:"), carbon copy lines ("Cc:" and "Bcc:"), a subject line ("Subject:") and an attachments line ("Attach:").
- A letter includes a letterhead, the date, an inside address, a greeting, and a closing with the writer's signature.
- A memo includes a memohead, the date, and lines for the reader ("To:"), the sender ("From:"), and the subject ("Subject:").

Figure 6.1 Formatting an E-mail, Letter, and Memo

E-mails (a), letters (b), and memos (c) are basically the same, except in their formatting. The main differences are that letters are written to readers outside the company, whereas memos are written to readers inside the company. E-mail can be written to readers both inside and outside the company.

(a)



The report is attached here as a pdf file.

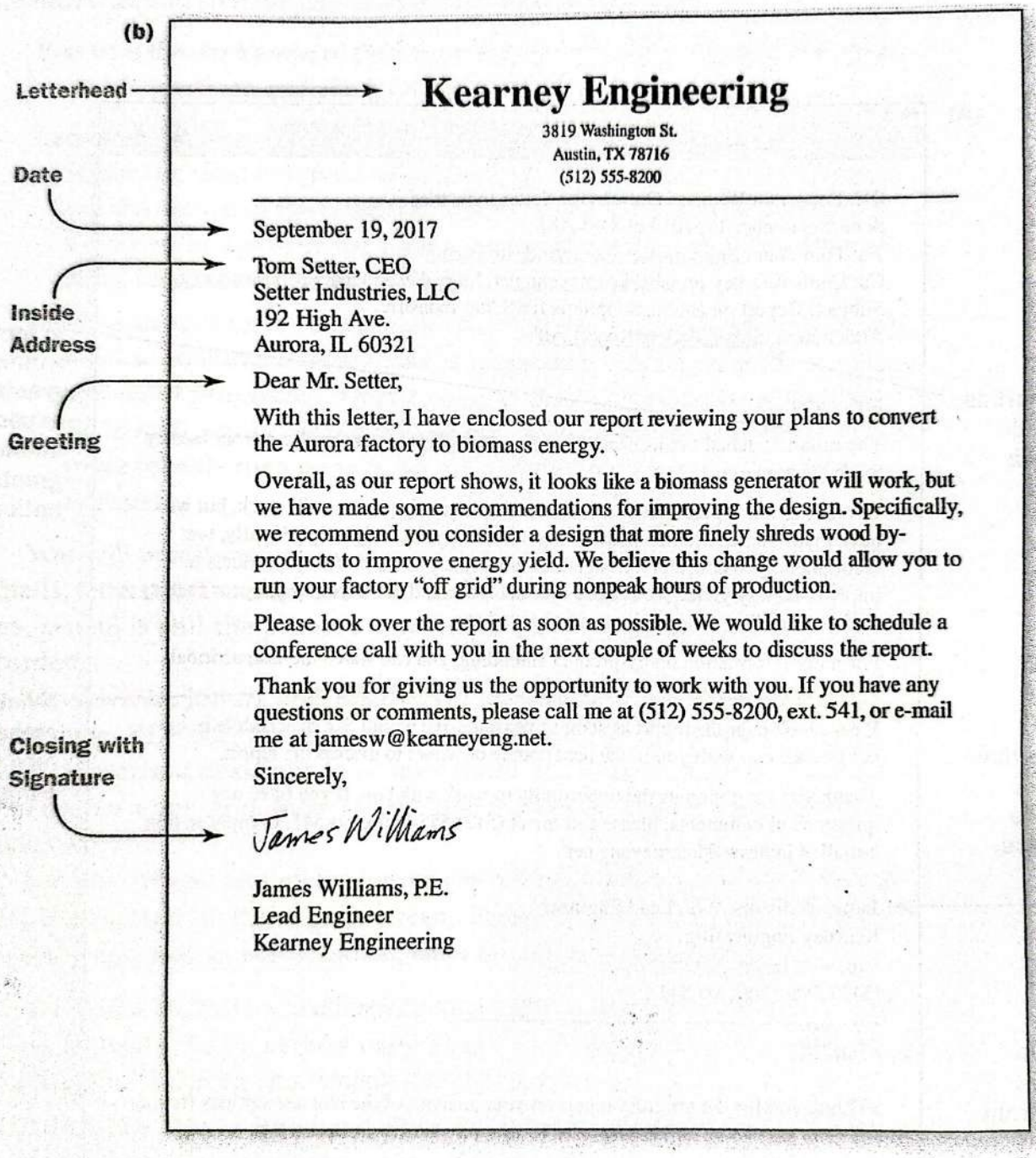
In formal e-mails, a greeting is expected. In informal e-mails, greetings are optional.

A link is embedded into the e-mail.

Here is the sender's premade signature.

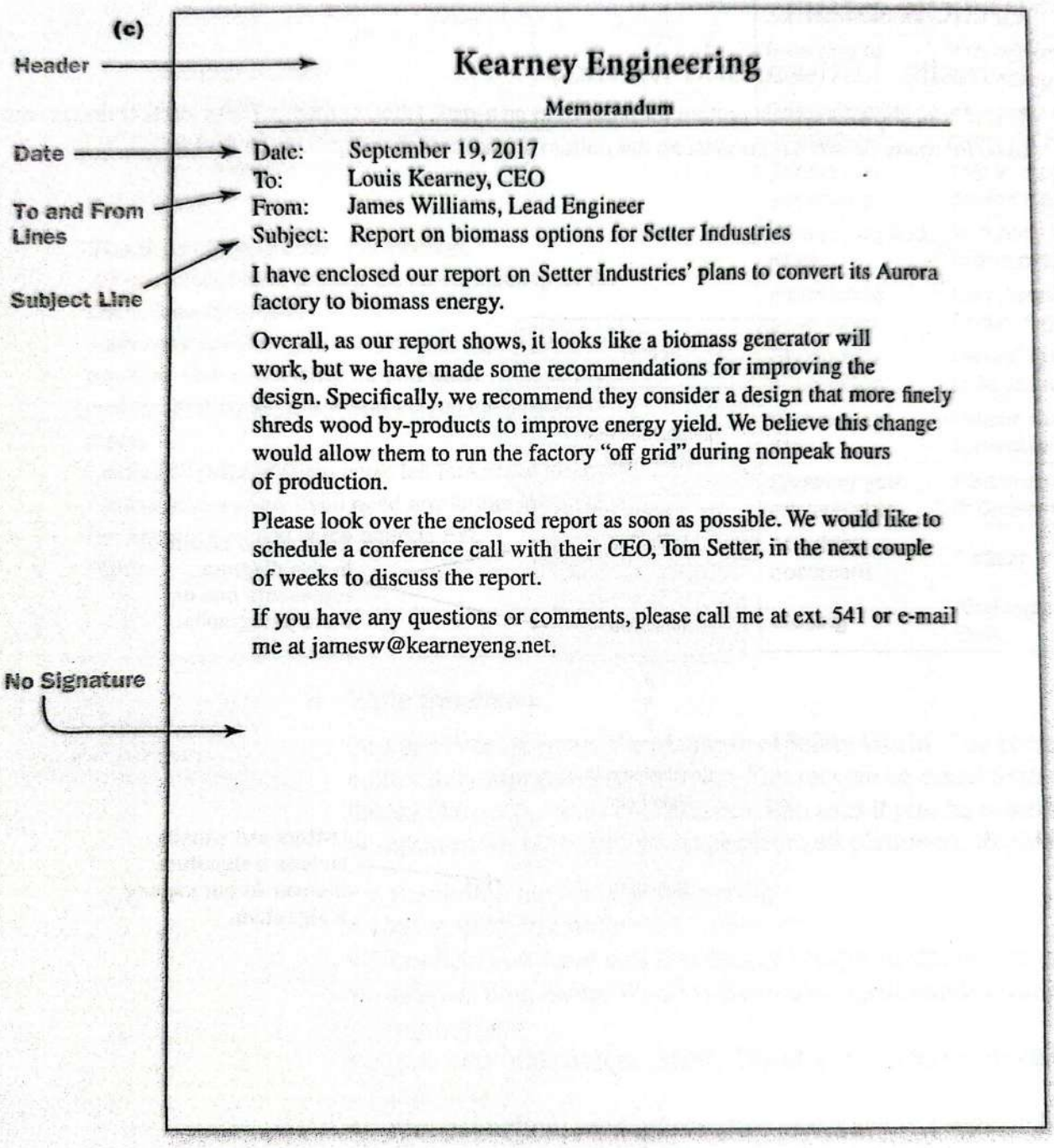
This is a previous e-mail from Tom Setter to which James is replying.

Figure 6.1 *(continued)*



(continued)

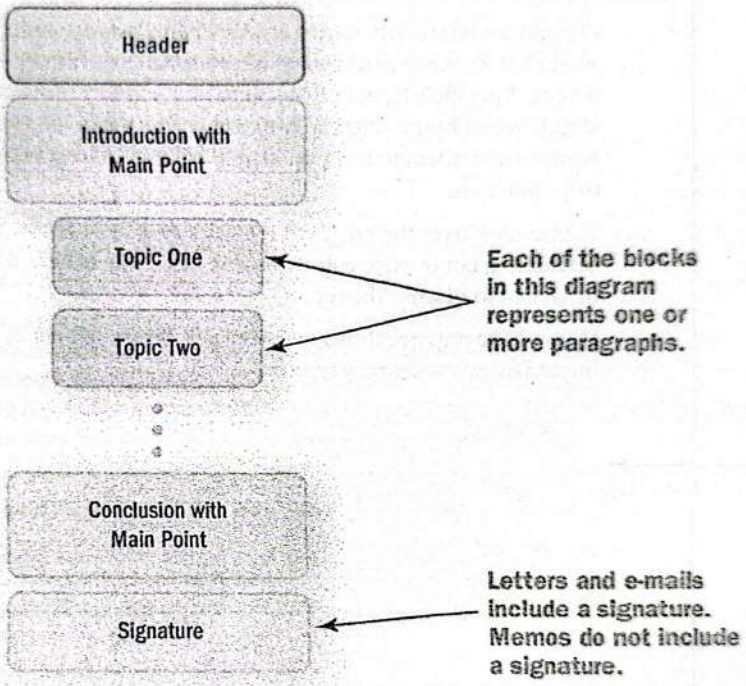
Figure 6.1 *(continued)*



Quick Start

E-mails, Letters, and Memos

This model shows a typical pattern for organizing an e-mail, letter, or memo. These kinds of documents are used for many different purposes, so the pattern shown here is designed to be flexible.



Basic Features of E-mails, Letters, and Memos

An e-mail, letter, or memo will generally have the following features:

- **Header** with the company name and address of the sender, as well as recipient's name and address
- **Greeting** or **salutation** for the recipient (not included in memos)
- **Introduction** that states a clear main point
- **Body paragraphs** that provide need-to-know information
- **Conclusion** that restates the letter's main point
- **Signature** of the sender (not included in memos)

Task **7** What things do you have to do today (or at the weekend)? Make a list, and then work out a timetable for doing them. Present your plan to the class.

Writing **8** Rewrite this email replacing the phrases in italics.

To: Crane Officer
From: Safety Officer
Subject: Change to safety meeting
Cc: Safety Manager

Thank you for your email this morning.
As you know, I have arranged a safety meeting for the Deck Crew tomorrow.
I am sorry to inform you that I cannot attend the meeting because I have sick leave for one week. However, *I can confirm that* my assistant Bob will run the meeting in my place.
I would be grateful if you could tell Tore about this change.
Please let me know if you need any further information.
I'm attaching a copy of the agenda FYI.
Rgds

Thanking for communication	*Thanks for; Many thanks for*
Referring to topic	*With reference to; With regard to; Concerning*
Reminding of background	*As you are aware,; As you may know,; As you may be aware,*
Confirming something	*This is to confirm that; I'd like to confirm that; I confirm that*
Introducing bad news	*I am sorry to tell you that; Unfortunately,*
Introducing good news	*I am pleased to inform you that; Fortunately,*
Giving new information	*I would like to inform you that; This is to let you know that*
Requesting action	*Please; Could you please; I would appreciate it if you could*
Showing you are available	*Please do not hesitate to contact me if; Do let me know if*
Attaching document	*I attach*
Closing	*Kind regards; Regards; Best wishes Best*

FYI = for your information

9 Write this email.

You are Pete Norman, the Manager of Safety World. The company makes and sells safety equipment for oil rigs. You receive an email from Ben Brenner, Safety Officer on Nord Oil Platform. Ben asks if you have any new safety equipment for emergency escapes from oil platforms. Reply to Ben.

- thank Ben for his email yesterday
- refer to his request
- confirm you have new emergency escape equipment for sale
- remind Ben: Safety World is Europe's largest manufacturer of oil safety equipment
- give new information: Safety World won the International Safety Award last year
- request action: send information about safety needs on Nord
- show you are available to answer any of Ben's questions
- attach photos of equipment in action

10 Exchange emails with a partner. Take the part of Ben, and reply to your partner's email.

- thank for email
- remind Pete: there was a serious fire on an oil platform last year; there were many casualties because of poor escape equipment
- give information about needs: you want new fire and emergency escape equipment such as heat shields, heat-resistant ropes, ladders, enough for crew of 230
- ask Pete to send catalogue for these items

3 Dealing with complaints

Start here

1 What are the most common customer complaints in your technical field?

2 How should staff deal with a complaint from a customer? Make some guidelines for staff (in note form).

Listening

3 22 Listen to this phone call from a customer with a complaint. Note down the details in the complaints form.

Date and time of call	10.45 16/03
Name of customer	
Order number	
Description of goods	
Model number	
Details of complaint	
Solution offered	☐ *replace* ☐ *repair* ☐ *refund* ☐ *reduce*
Customer response	☐ *accept* ☐ *reject*

4 Listen again to the phone call. Look at the company handbook below. Which procedures are (a) ***followed*** (b) ***broken*** by the staff?

Procedure for dealing with a telephone complaint from a customer

1. be friendly, polite and helpful
2. listen carefully
3. show sympathy with the customer's problem, but don't admit the company's fault
4. summarise what the customer has told you, and check that you have understood correctly
5. record the details and collect the evidence (e.g. receipts or damaged goods)
6. offer a solution (*repair* the item, *replace* it, *refund* the money or *reduce* the price of the next purchase)

Task

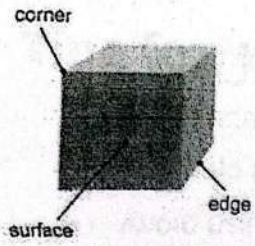
5 Work in pairs, A and B. Roleplay a phone call between service staff and a customer with a complaint. Before you start, study the audio script of the phone call in 3 on page 124.

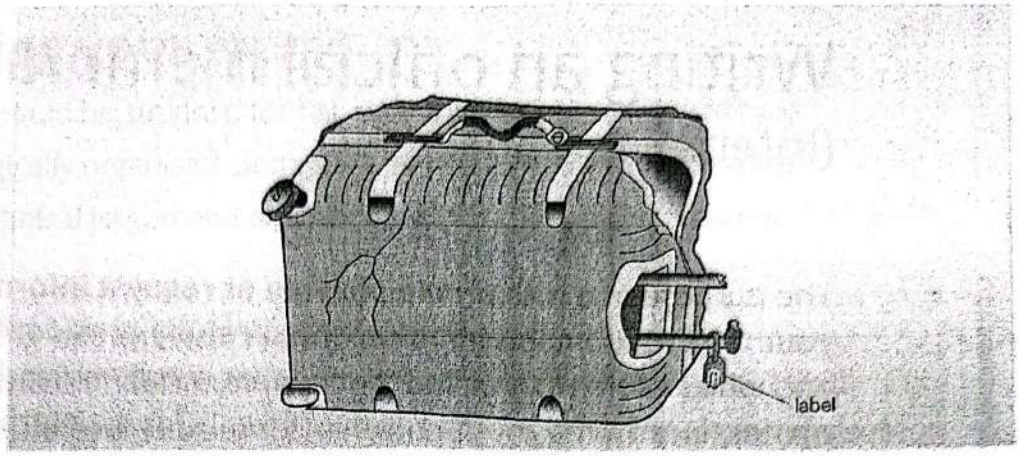
Student A. You're the customer. Make notes about your device and what is wrong with it. Include a model name, number and an order number. Then call customer services.

Student B. You work in customer service. Draw up a customer complaints form like the one in 3. Then take the call from the customer and deal with their complaint.

6 Describe the damage to the suitcase.

broken burnt cracked crushed dented split torn twisted





Reading **7** Read this reply to a customer's letter of complaint and answer the questions.

Dear Ms Beck

Thank you for your letter complaining about your DVD player from our store. I was sorry to hear that the top surface of the machine was scratched and the edge was cracked.

Unfortunately, we do not have any more DVD players in stock at the moment. However, I am pleased to inform you that we will give you your money back in full. In addition, as a gesture of goodwill, we will give you a 10% discount off your next purchase from our store.

I would like to apologise for the inconvenience you have experienced. Please do not hesitate to contact me if you have any further queries.

Yours sincerely

Robert Wilson

Store Manager

1 What did the customer complain about?
2 What does the writer offer to do?
3 Which words (a) show sympathy (b) give an apology?
4 Which words introduce (a) good news (b) bad news?

Writing **8** You are the Manager of IT Online Ltd. Reply to this letter.

The Manager, IT Online Ltd

Dear Sir or Madam

I wish to complain about the Mace notebook computer which I ordered from your online store last month (order number 60335/01). The computer was delivered only yesterday. When I opened the box, the adapter was missing. This is very poor service indeed. I look forward to hearing from you.

Peter Bradwell

Writing an official memorandum
(Internal memo)

The purpose of a memo is **to provide or request information** from people within your organisation. It is important to **carefully consider the key point** of the memo and who should **receive** it. A memo that is written clearly **using a friendly and professional tone** avoids any opportunity for misinterpretation.

Structure of a memo

Part 1: HEADER
TO: provide the names and titles of everyone who will receive your memo
FROM: provide your complete name and title
DATE: provide the complete and accurate date – don't forget to include the year
SUBJECT: provide a brief, yet specific description of what the memo is about

Part 2: MESSAGE
Introduction – explain the purpose
Body – addresses the information required
Conclusion – (Summary and possible recommendations)

Presentation/Style:

- is Word-processed
- has a simple and professional look
- is centred on the page from top to bottom
- is left justified, with single or 1.5 spacing within paragraphs and double spacing between paragraphs
- uses a 12-point font size and a standard font like Times New Roman or Arial that is easy to read
- includes sufficient white space to enhance clarity

Tips for writing your memo

- Your memos should be succinct, formal, clear, interesting and easy to read
- It should be logically organised, accurate, well-researched and informative
- Avoid using technical jargon and abbreviations that the recipient may not understand
- Avoid the use of slang, colloquialisms and contractions
- Use action verbs and an active voice to convey your confidence
- Highlight important points and reduce unnecessary wordiness with headings, sub-headings, bullet points and numbered lists
- Check for correct grammar, spelling and punctuation

Business Memo Template Format

Header:

In your header, you'll want to clearly label your content "Memorandum" so your readers know exactly what they're receiving. Then, you'll want to include "TO", "FROM", "DATE", and "SUBJECT". This information is relevant for providing content, like who you're addressing, and why.

Paragraph One:

In the first paragraph, you'll want to quickly and clearly state the purpose of your memo. You might begin your sentence with the phrase, "I'm writing to inform you ... " or "I'm writing to request ... ". A memo is meant to be short, clear, and to the point. You'll want to deliver your most critical information upfront, and then use subsequent paragraphs as opportunities to dive into more detail.

Paragraph Two:

In the second paragraph, you'll want to provide context or supporting evidence. For instance, let's say your memo is informing the company of an internal re-organization. If this is the case, paragraph two should say something like, "As our company continues to grow, we've decided it makes more sense to separate our video production team from our content team. This way, those teams can focus more on their individual goals."

Paragraph Three:

In the third paragraph, you'll want to include your specific request of each employee — if you're planning a team outing, this is the space you'd include, "Please RSVP with dietary restrictions," or "Please email me with questions."

On the contrary, if you're informing staff of upcoming construction to the building, you might say, "I'd appreciate your cooperation during this time." Even if there isn't any specific action you expect from employees, it's helpful to include how you hope they'll handle the news and whether you expect them to do something in response to the memo.

Memo Example 1: A General Office Memo

MEMORANDUM

To: All Staff

From: The Manager

Date: May 27, 2021

Subject: Inappropriate use of time on Google Doodle games

Coworkers,

It has come to my attention that many in the office have been spending time on the Google home page microgames. This memo is a reminder to use your work hours for work.

According to a recent article, the estimated daily cost of people collectively playing these games instead of working is over $120 million—which is calculated based on the daily average increased time spent on the Google home page (36 seconds).

If these estimates are applied to our 600 office employees, this results in a nearly $700 weekly loss.

This is a conservative estimate considering the extensive discussions that occur about beating the office's current high score. The extra cost quickly adds up.

Of course, we don't want you to view our organization as a place of drudgery and draconian rules. I encourage a fun and competitive environment, and I recognize that we certainly won't be profitable if you are unhappy or dissatisfied with your jobs. This is just a reminder to be careful with your use of company time.

Thank you,

The Manager

Memo Example 2: A Departmental Memo

MEMORANDUM

To: Computer Programming Division

From: Vice President Lumbergh

Date: February 19, 2021

Subject: Attaching cover sheets to TPS reports

This is to remind the division that, starting today, we are now filing all Testing Procedure Specification (TPS) reports with new cover sheets.

The reason for this change is simple. In addition to a new format, the cover sheets provide a summary of the report as well as the updated legal copy. The new cover sheets also include Initech's new logo.

Though this change may initially seem like a headache and an extra step, it is necessary to include the new cover sheets due to their updated information. Failing to do so will result in a confusing and inaccurate product being delivered to our customers.

Please be sure to follow this new procedure.

Best regards,

Vice President Lumbergh

Reference: https://www.learninghub.ac.nz/writing/writing/writing-a-memorandum/

Manalo, E., Wong-Toi, G. & Bartlett-Trafford, J. (2009). The business of writing (3rd ed.). New Zealand: Pearson Education New Zealand.

Elements of Inquiry Letter

An inquiry letter sample of this kind must have the following pieces of information:

Names and Addresses of the Parties – You have to incorporate the names and the addresses of the two parties involved. The two parties here are you, the sender, and the recipient (organization involved).

Subject Line – It is here that you drop a one-line explanation of the nature of the inquiry. The purpose of this line is to raise the curiosity of the reader while at the same time prepare him for the body of the sample inquiry letter.

Introduction – The introduction is a one-paragraph overview of the problem at hand. It simply states the nature of the inquiry and what exactly you are looking at. You may introduce yourself briefly as well.

Justification for the Seeking – This one extends from the introduction and goes to explain exactly why the information, item or property is sought. Needless to say, the reasons have to be truly valid for them to be deemed valid. Write precisely. Use one-two paragraphs for explanation.

Supportive Affidavits – To up your chances of sailing through, you have to attach supportive affidavits like statements from persons of authorities, official letters, writing samples, and other forms of references.

Signature and Date – As with any other official letter, this one also must bear the relevant signature and date. These unique identifiers have to be placed right at the bottom of the inquiry letter. Thank them before you put your name and signature.

Inquiry Letter for an English course

Date:

_____ (Name),
_____ (Address)

Subject: Enquiry about English course

Dear Sir/Madam/Name,

My name is _____ (Name), and I am pursuing _____ (Course) from _____ (School/ College Name). I am writing this letter to you in order to enquire about the English course that you offer.

I would like to state that I am looking for an English course that would be helping me in the coming future with applying in foreign countries. Your institute is reputed and well known for good courses. Therefore, I am willing to apply to your institute.

I kindly request you to inform me about the charges, duration, class timings, working days and all other relevant information. Please guide me through the admission and other procedures. I shall be highly thankful.

I look for your kind response. You may contact me at _____.

Best regards,
_____ (Name),
_____ (Contact Number)

Inquiry Letter for Computer

17 June 2022

Robert Brown
65H Dragan
Tucson, AZ, 85705

Subject: Inquiry about a computer

Dear Mr. Lloyd,

This letter is to inquire about the Lenovo Laptop. I am planning to buy fifteen laptops for my business purposes.

We are an advertising and public relations firm and would require a laptop with graphic features. The fifteen laptops' budget is seven lakhs; We would need laptops that are easy to carry, portable, have a perfect ram size, and, most importantly, meets our budget.

I would like to know about the varieties of laptops that fit the needs I have mentioned above. If another brand matches the requirements, I am okay with that too.

Please do not hesitate to contact me at mlisa@gmail.com if you have any questions.

Thank you for your time.

Regards,

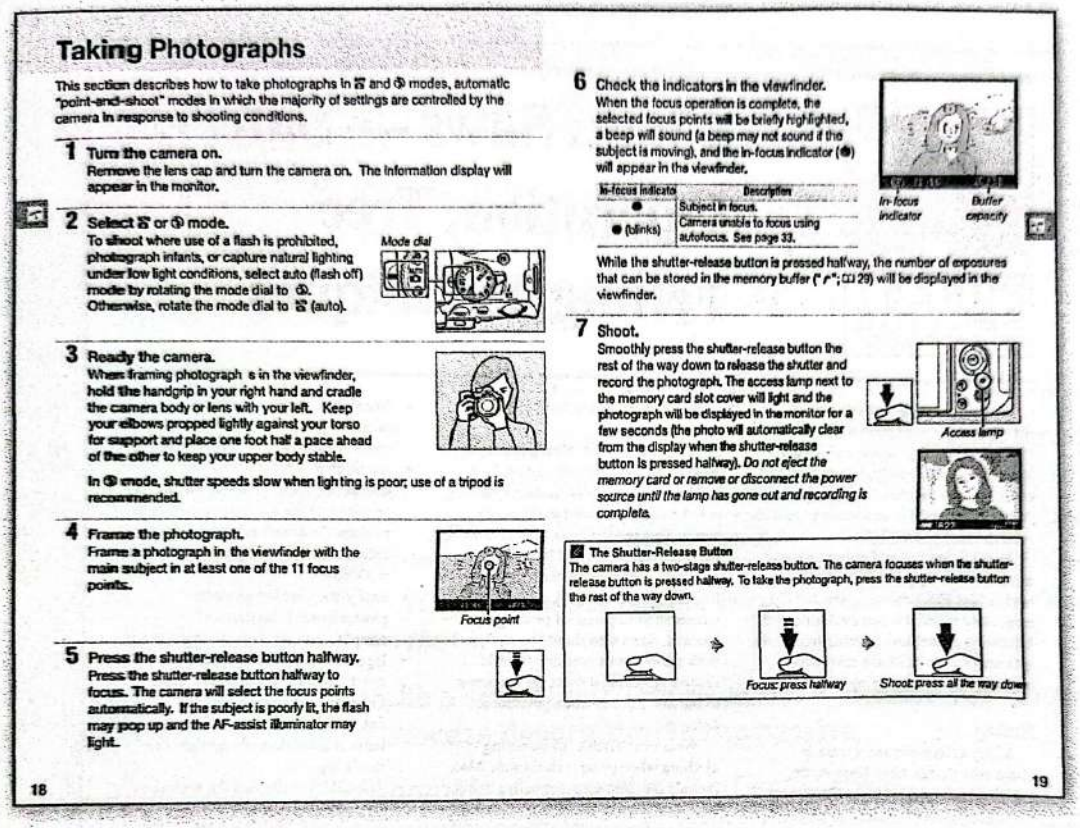
Mary Lisa
Managing Director
Brookes IT Limited

Writing Instructions & Description of process

Figure 8.7 Instructions with Sequentially Ordered Steps

Each step should express one action. Putting the steps in list format makes them easier to follow.

SOURCE: Nikon D5100 User Manual, pages 10 and 11. © 2014 Nikon Corporation. Courtesy of Nikon Inc., Melville, New York.



Taking Photographs

This section describes how to take photographs in 📷 and Ⓢ modes, automatic "point-and-shoot" modes in which the majority of settings are controlled by the camera in response to shooting conditions.

1 Turn the camera on.
Remove the lens cap and turn the camera on. The information display will appear in the monitor.

2 Select 📷 or Ⓢ mode.
To shoot where use of a flash is prohibited, photograph infants, or capture natural lighting under low light conditions, select auto (flash off) mode by rotating the mode dial to Ⓢ. Otherwise, rotate the mode dial to 📷 (auto).

Mode dial

3 Ready the camera.
When framing photographs in the viewfinder, hold the handgrip in your right hand and cradle the camera body or lens with your left. Keep your elbows propped lightly against your torso for support and place one foot half a pace ahead of the other to keep your upper body stable.

In Ⓢ mode, shutter speeds slow when lighting is poor; use of a tripod is recommended.

4 Frame the photograph.
Frame a photograph in the viewfinder with the main subject in at least one of the 11 focus points.

Focus point

5 Press the shutter-release button halfway.
Press the shutter-release button halfway to focus. The camera will select the focus points automatically. If the subject is poorly lit, the flash may pop up and the AF-assist illuminator may light.

6 Check the indicators in the viewfinder.
When the focus operation is complete, the selected focus points will be briefly highlighted, a beep will sound (a beep may not sound if the subject is moving), and the in-focus indicator (●) will appear in the viewfinder.

In-focus indicator	Description
●	Subject in focus.
● (blinks)	Camera unable to focus using autofocus. See page 33.

In-focus indicator Buffer capacity

While the shutter-release button is pressed halfway, the number of exposures that can be stored in the memory buffer ("r"; 29) will be displayed in the viewfinder.

7 Shoot.
Smoothly press the shutter-release button the rest of the way down to release the shutter and record the photograph. The access lamp next to the memory card slot cover will light and the photograph will be displayed in the monitor for a few seconds (the photo will automatically clear from the display when the shutter-release button is pressed halfway). *Do not eject the memory card or remove or disconnect the power source until the lamp has gone out and recording is complete.*

Access lamp

The Shutter-Release Button
The camera has a two-stage shutter-release button. The camera focuses when the shutter-release button is pressed halfway. To take the photograph, press the shutter-release button the rest of the way down.

Focus: press halfway Shoot: press all the way down

18 19

STATE ONE ACTION PER STEP Each step should express only one action (Figure 8.7). You might be tempted to state two smaller actions in one step, but your readers will appreciate following each step separately.

Ineffective

2. Place the telescope securely on its side as shown in Figure 4 and open the battery compartment by simultaneously depressing the two release latches.

Revised

2. Place the telescope securely on its side as shown in Figure 4.
3. Open the battery compartment by simultaneously depressing the two release latches.

However, when two actions must be completed at the same time, you should put them in the same sentence.

6. Insert a low-power eyepiece (e.g., 26mm) into the eyepiece holder and tighten the eyepiece thumbscrew.

Figure 8.8 Paragraph Style Instructions

Paragraph style often takes up less space. It is a little harder to read at a glance, but this style can often be more personal. In these instructions, the pictures show helpful details that clarify the steps.

SOURCE: "Chain Saws—Safety, Operation, Tree Felling Techniques," by Eric Ward, Kansas Forest Service. Published by Kansas State University, Manhattan, KS, April 2011, MF2013Rev.

Paragraphs slow the readers down.

Figures are numbered and referenced in the written text.

Drawings are used to illustrate complex steps.

Safety equipment is used in drawings.

KANSAS FOREST SERVICE
KANSAS STATE UNIVERSITY

Chain Saws — Safety, Operation, Tree Felling Techniques

A chain saw is a valuable, labor saving tool for homeowners, forest landowners, and professional loggers. When used improperly, however, it can cause serious injuries. Read and follow operating manual instructions provided with the chain saw.

Good judgment and common sense are essential to operating a chain saw safely. Equipment varies, but if a chain saw manual is not available, the following guidelines provide important information about chain saw safety, operation, and tree felling techniques.

Safety

Many safety advancements in chain saw design have been made, but accidents still happen. Chain saw users must observe safety practices. Working safely requires a personal commitment to being constantly aware of your actions and the possible reactions they may cause, as well as your surroundings.

Select a chain saw with good safety features, including features to reduce kickback and an anti-vibration system to reduce saw vibration to the user's hands. This reduces user fatigue and ensures greater safety. Chain saw operators should know their physical limitations, work slowly, rest often, and remain alert to potential problems and hazards. Saws also should be equipped with a continuous pressure throttle control system that shuts off the power to the saw's chain when pressure is reduced.

Wear comfortable, close-fitting clothing when using a chain saw. Also, include the following protective equipment (see Figure 1):

- safety boots with steel toes and nonskid soles.
- face shields or plastic goggles to prevent injuries from wood chips and sawdust.
- ear plugs or muffs to prevent hearing loss. Chain saw noise is greater than the human ear can tolerate. Sustained exposure can cause hearing loss that cannot be restored.
- heavy-duty, leather gloves to protect hands from cuts and scrapes.
- leg protective pants or chaps that cover the upper thigh to boot tops. Chaps protect against cuts and can stop the chain before it causes harm if it accidentally contacts the user's leg.
- first-aid kit available at the work site.

Figure 1. *Chain saw operator's safety equipment.*

Before starting the chain saw, check the operating manual for the recommended fuel mixture, choke setting, and throttle control. Always start the saw with the chain brake on. Properly adjust the saw so the chain stops when the throttle is released. Start the chain saw at least 10 feet away from the fueling area. Always fill the oil reservoir when refueling the saw.

When starting the saw, hold the saw firmly on the ground with your right foot in the rear handle. Grip the front handle with the left hand. Be sure the area under the bar and chain is clear. Check to see that the starter mechanism is engaged, then pull the starter rope sharply with the right hand while keeping a firm hold on the

Kansas Forest Service

K-STATE

Language In an active sentence, the subject = the agent. The subject does the action.

Subject = agent	Active verb	Object
Hydraulic cylinders	push	the cutter.
Large scoops	collect	the rocks.

In a passive sentence, the subject is NOT the same as the agent. The subject does not do the action. The agent does the action to the subject.

Subject	Passive verb		Agent
	be	Past participle	
The cutter	is	pushed	by hydraulic cylinders.
The rocks	are	collected	by large scoops.

6 Change this set of instructions into a description of a process, using the passive and the words in the box.

finally first next now then

How to change the oil in a car

1	Run the engine for a few minutes.	5	Put the oil drain plug on
2	Switch off the engine.	6	Take off the oil filler cap.
3	Take off the oil drain plug.	7	Pour in the new oil.
4	Empty the old oil into a container.	8	Put the oil filler cap back on.

Begin: First the engine is run for a few minutes. Then it is switched off. Now the ...

7 Make a set of instructions about a process you know about. Then rewrite it as a process description in the passive.

Examples of processes: food manufacture, steel making, canning, assembling computer components, manufacturing a CD, dairy processing.

8 Fill in the gaps, using the correct form of the verbs in brackets.

1 Large drills _____ (make) holes in the roof of the tunnel. Then the holes _____ (fill) with bolts and cement.
2 A large propeller _____ (push) the hovercraft forwards. The propeller _____ (drive) by a powerful engine.
3 Hot water _____ (flow) from the engine into the radiator. Here it _____ (cool) by the fan.
4 The robot _____ (monitor) by a computer. This computer also _____ (control) all the other robots in the building.
5 First, the rusty machine parts _____ (bring) into the factory. Then they _____ (clean). Then the rust _____ (remove). Next the parts _____ (paint). Finally, they _____ (take) out of the factory again.

9 Make a list of headings for the main stages of a process you know about. Make each heading begin with a verb ending in ***-ing***, like the ones in 3.

Example: Moulding and shaping steel – 1 Melting the steel; 2 Casting; 3 Cooling; 4 Rolling the steel; 5 Straightening; 6 Cutting.

10 Give a short talk to the class explaining your process. Use your headings.

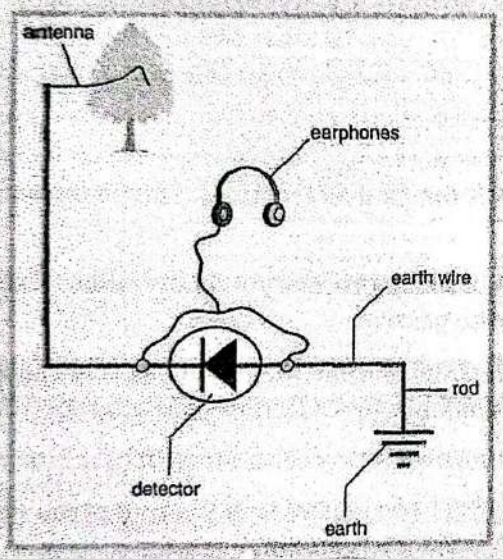
8 Change these instructions into a description of a process, using the passive.

How to clean a spark plug

1 Take off the spark plug cover.
2 Loosen the spark plug with a special wrench.
3 Remove the spark plug from the socket.
4 Clean the spark plug using a wire brush.
5 Replace the spark plug in the socket.
6 Tighten the spark plug using the wrench.
7 Put the cover back on the spark plug.

Begin: First of all, the spark plug cover is taken off. Then the spark plug is ...

9 Change the second paragraph into a set of instructions, using imperatives.



You can make your own radio using a few simple components: two lengths of wire (one 3 m long, and the other 6 m long), a metal rod, earphones and a detector. This is how to do it.

First, the rod is hammered into the ground. Then the insulation is stripped off the end of the 3-metre wire. The wire is twisted around the rod ten times to make a good connection. This is the earth wire. Next, the detector is attached to the other end of the earth wire. The 6-metre wire is now taken and one end is connected to the other end of the detector. (This wire is your antenna.) The antenna is hung from a tree (making sure that the bare end does not touch the earth). The two wires from the earphones are connected to each end of the detector. Finally, the earphones are put on. Now you can hear the radio station (if you are very close to the transmitter!).

Begin: 1 Hammer the rod into the ground.
2 Strip the insulation off the end of the 3-metre wire.

10 Make a set of headings for a talk on these topics. Make each heading begin with a verb ending in ***-ing***.

1 First, I'd like to talk about how the communications satellite is launched.
2 After that, I'll talk about how the programmes are transmitted to the satellite.
3 Then I'll look at how the digital signals are received from the satellite.
4 Next, I'll explain how your satellite dish and digital receiver are installed.
5 Then I'll go on to mention how your dish is connected to the digital TV receiver.
6 The next topic is how high-frequency signals are converted to low-frequency ones.
7 And then I'll move on to how the video and audio are extracted from the digital signal.
8 Finally, I'll mention how the video and audio are played via the monitor and speakers.

Example: 1 Launching the communications satellite

11 Complete these. Use hyphens (-). Note: Be careful with plural nouns.

1 The plane is ready for the road. It's a *road-ready* plane.
2 The engine has a cycle of four strokes. It's a *4-stroke* engine.
3 The propeller has three blades. It's a _____ propeller.
4 The cable is six metres long. It's a _____ cable.
5 This computer is activated when you use your voice. It's a _____ computer.
6 That ticket machine starts when you touch the screen. It's a _____ ticket machine.

12 Ask and answer questions about a car assembly plant.

location = where something happens
destination = where something is going to

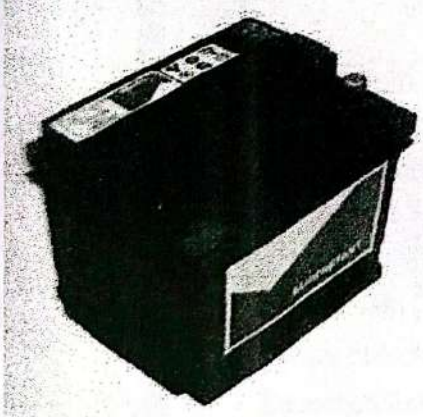
	Action	Purpose, method, agent, time, location, destination
1	deliver car parts	a) *method*: truck or rail b) *destination*: delivery area
2	carry parts	a) *destination*: different parts of plant b) *method*: forklift trucks or conveyor belts
3	weld panels to frame	a) *location*: body shop b) *agent*: 400 robots c) *purpose*: make the body of car
4	check the car body	a) *time*: after painting b) *agent*: human workers c) *purpose*: look for faults in the paint
5	insert windscreen	a) *destination*: front of car body b) *agent*: robots c) *method*: laser guides
6	move chassis and body simultaneously	a) *destination*: final assembly line b) *purpose*: attach body to chassis

1 a) How are the car parts delivered? They're delivered by truck or rail.
b) Where are they delivered? To the delivery area.

13 Write full sentences using the passive.

Example: 1 The car parts are delivered to the delivery area by truck or rail.

14 Rewrite this set of instructions as a paragraph describing a process. Use the passive form of the verbs.



Servicing a car battery

Open the bonnet of the car. Locate the battery.
Loosen the battery cables, using a wrench. Remove the battery cables from the posts.
Always remove the negative (or earth) cable first, then the positive.
Carefully lay the detached ends of the cables to one side.
Wipe away corrosion from the top of the battery, using baking soda and water.
If corrosion is very heavy, you can clean it from the posts using a wire brush.
Apply petroleum jelly to the inside of the terminals and the posts.
Reattach the cables. Close the car bonnet.

Begin: First the bonnet of the car is opened and the battery is located. Then ...

Project **15** Research an industry you are interested in.

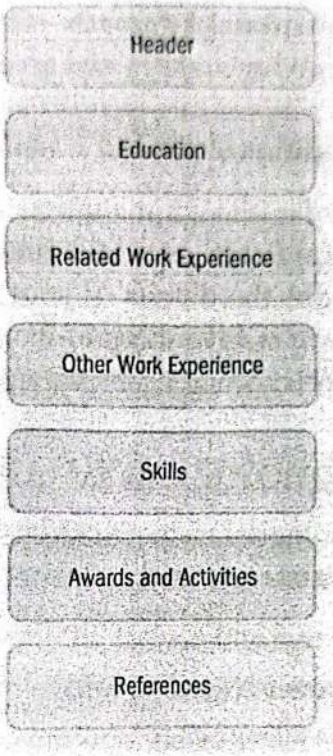
- Find out about an important process in the industry.
- Draw a flow chart of the main stages in the process.
- Write a description of the process.
- Explain the process to the class.

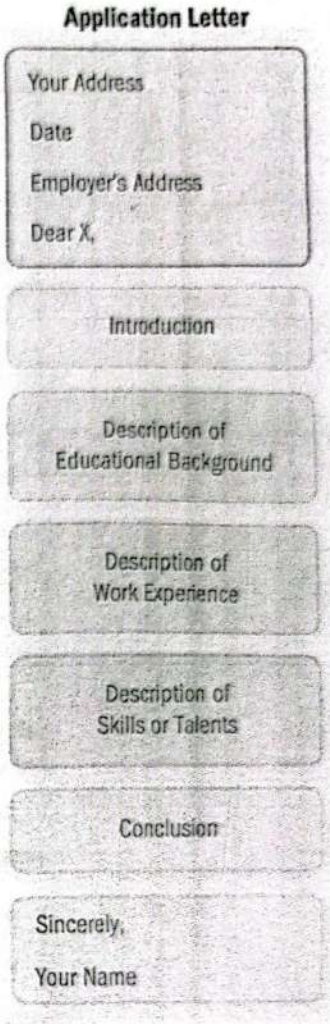
Writing Resume & Cover Letter

Quick Start

Career Materials

These models show typical organizational patterns for a résumé and application letter. These patterns should be altered to fit your background and the kinds of jobs you are applying for.





Basic Features

Minimally, your career materials will include a résumé and an application letter. These documents will usually have the following features:

Résumé

- Header with your name and contact information
- Career objective/career summary
- Educational background that lists your degrees and training
- Work experience in your field of study
- Other work experience not in your field of study
- Skills related to the career you are pursuing
- Awards you have earned and activities in which you have participated
- References listing employers, professors, and other professionals

Application Letter

- Header with addresses
- Introduction that identifies position being applied for
- Description of your educational background
- Description of your work experience
- Description of your specialized skills or talents
- Conclusion that thanks readers

Types of Résumés

Résumés tend to follow one of two organizational approaches: the *chronological* approach and the *functional* approach.

Chronological approach—Organizes the résumé according to education and work experience, highlighting a job seeker's qualifications in a few areas. A chronological résumé might be organized into sections such as Education, Work Experience, Skills, and Awards/Activities.

Functional approach—Organizes the résumé according to your talents, abilities, skills, and accomplishments. A functional résumé might be organized into sections such as Leadership, Design Experience, Communication Skills, and Training Abilities.

By far, the chronological approach is the most common in scientific and technical fields, especially for new college graduates, because it allows you to highlight the details of your education and qualifications. The functional approach is advantageous for more experienced job seekers who want to highlight their accomplishments and experience.

Chronological Résumé

Chronological résumés can be divided into the following sections, which can be organized in a variety of ways:

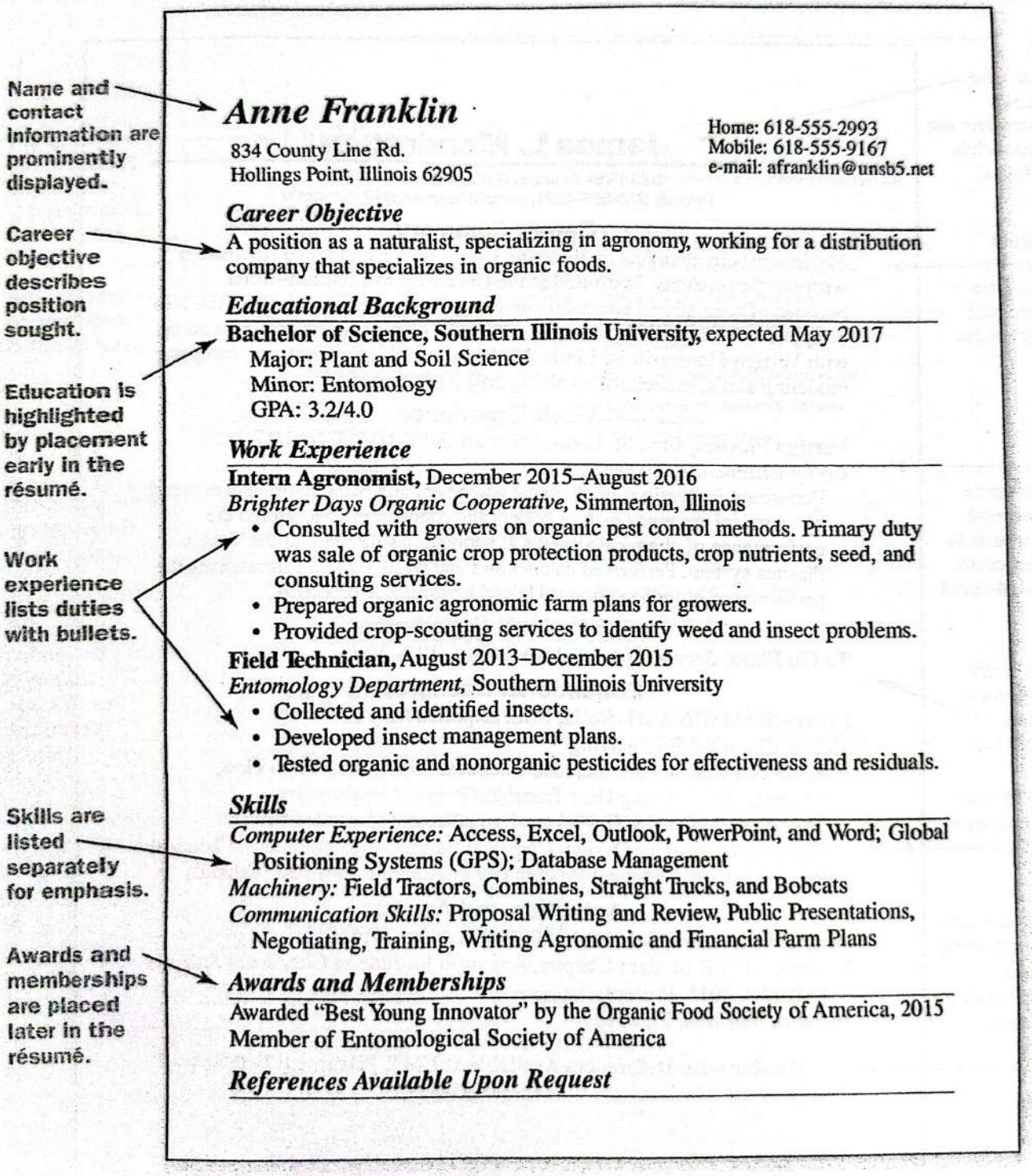
- Name and contact information
- Career objective/career summary
- Educational background
- Related work experience
- Other work experience
- Skills
- Awards and activities
- References

The headings in this list are common in chronological résumés, but other headings are available. The résumés shown in Figures 5.1, 5.2, 5.3, and 5.4 show how these headings can be varied, depending on your background and career path.

NAME AND CONTACT INFORMATION At the top of the page, your résumé should include a heading that states your name, address, phone number, and e-mail address. To catch the reader's eye, your name should appear in a larger font size. The heading can be left justified, centered, or right justified, depending on the design of your résumé.

Figure 5.1 Chronological Résumé

Anne Franklin's résumé uses a list style. She achieves a classic look by using a serif font (Garamond) and left justification.



Name and contact information are prominently displayed.

Career objective describes position sought.

Education is highlighted by placement early in the résumé.

Work experience lists duties with bullets.

Skills are listed separately for emphasis.

Awards and memberships are placed later in the résumé.

Anne Franklin

834 County Line Rd.
Hollings Point, Illinois 62905

Home: 618-555-2993
Mobile: 618-555-9167
e-mail: afranklin@unsb5.net

Career Objective

A position as a naturalist, specializing in agronomy, working for a distribution company that specializes in organic foods.

Educational Background

Bachelor of Science, Southern Illinois University, expected May 2017
Major: Plant and Soil Science
Minor: Entomology
GPA: 3.2/4.0

Work Experience

Intern Agronomist, December 2015–August 2016
Brighter Days Organic Cooperative, Simmerton, Illinois

- Consulted with growers on organic pest control methods. Primary duty was sale of organic crop protection products, crop nutrients, seed, and consulting services.
- Prepared organic agronomic farm plans for growers.
- Provided crop-scouting services to identify weed and insect problems.

Field Technician, August 2013–December 2015
Entomology Department, Southern Illinois University

- Collected and identified insects.
- Developed insect management plans.
- Tested organic and nonorganic pesticides for effectiveness and residuals.

Skills

Computer Experience: Access, Excel, Outlook, PowerPoint, and Word; Global Positioning Systems (GPS); Database Management
Machinery: Field Tractors, Combines, Straight Trucks, and Bobcats
Communication Skills: Proposal Writing and Review, Public Presentations, Negotiating, Training, Writing Agronomic and Financial Farm Plans

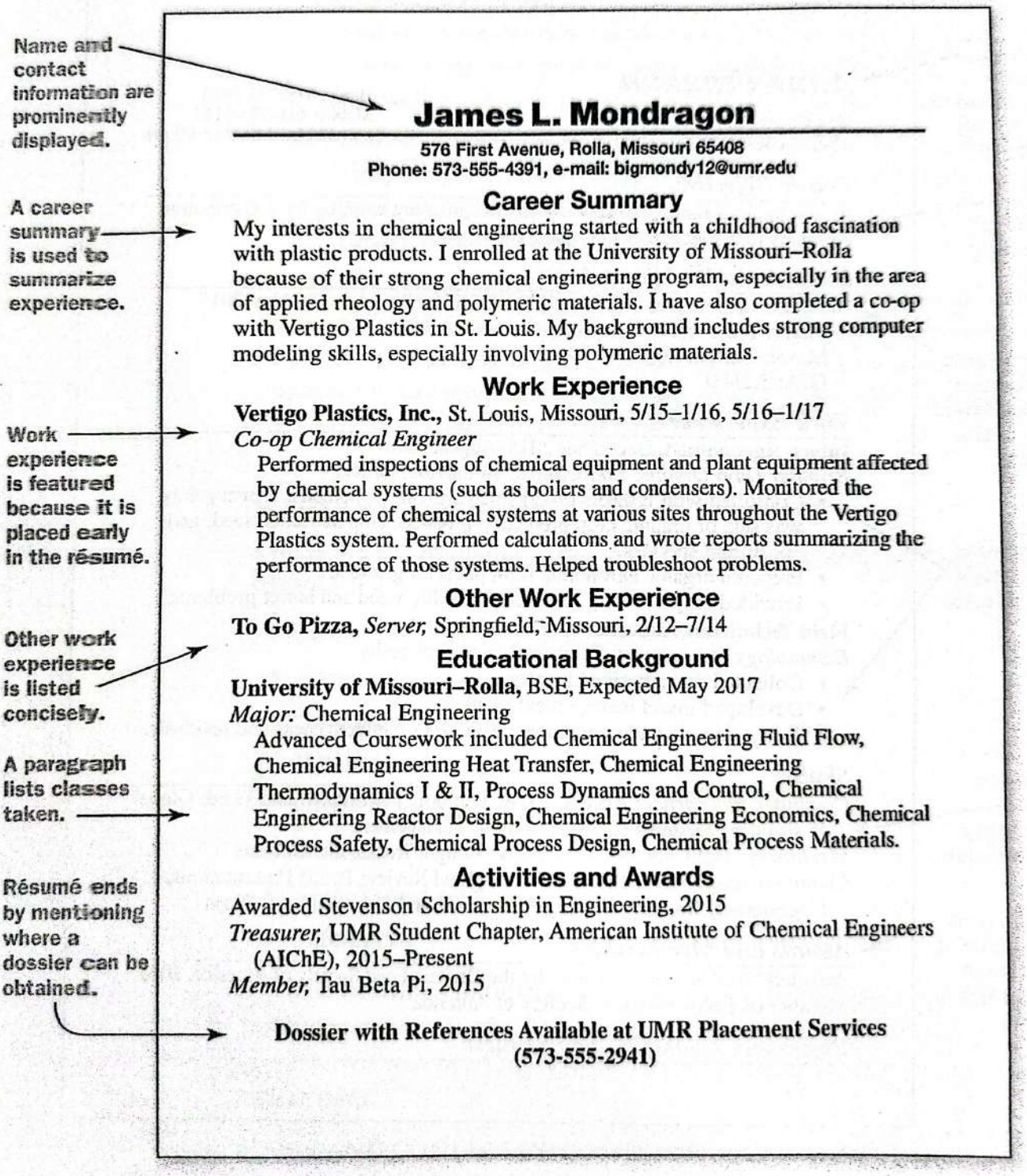
Awards and Memberships

Awarded "Best Young Innovator" by the Organic Food Society of America, 2015
Member of Entomological Society of America

References Available Upon Request

Figure 5.2 Chronological Résumé

James Mondragon's résumé uses a paragraph style. It uses a sans serif font for headings (Myriad) and a serif font (Times) for the body text. The headings are centered.



Name and contact information are prominently displayed.

A career summary is used to summarize experience.

Work experience is featured because it is placed early in the résumé.

Other work experience is listed concisely.

A paragraph lists classes taken.

Résumé ends by mentioning where a dossier can be obtained.

James L. Mondragon

576 First Avenue, Rolla, Missouri 65408
Phone: 573-555-4391, e-mail: bigmondy12@umr.edu

Career Summary

My interests in chemical engineering started with a childhood fascination with plastic products. I enrolled at the University of Missouri–Rolla because of their strong chemical engineering program, especially in the area of applied rheology and polymeric materials. I have also completed a co-op with Vertigo Plastics in St. Louis. My background includes strong computer modeling skills, especially involving polymeric materials.

Work Experience

Vertigo Plastics, Inc., St. Louis, Missouri, 5/15–1/16, 5/16–1/17
Co-op Chemical Engineer
Performed inspections of chemical equipment and plant equipment affected by chemical systems (such as boilers and condensers). Monitored the performance of chemical systems at various sites throughout the Vertigo Plastics system. Performed calculations and wrote reports summarizing the performance of those systems. Helped troubleshoot problems.

Other Work Experience

To Go Pizza, *Server,* Springfield, Missouri, 2/12–7/14

Educational Background

University of Missouri–Rolla, BSE, Expected May 2017
Major: Chemical Engineering
Advanced Coursework included Chemical Engineering Fluid Flow, Chemical Engineering Heat Transfer, Chemical Engineering Thermodynamics I & II, Process Dynamics and Control, Chemical Engineering Reactor Design, Chemical Engineering Economics, Chemical Process Safety, Chemical Process Design, Chemical Process Materials.

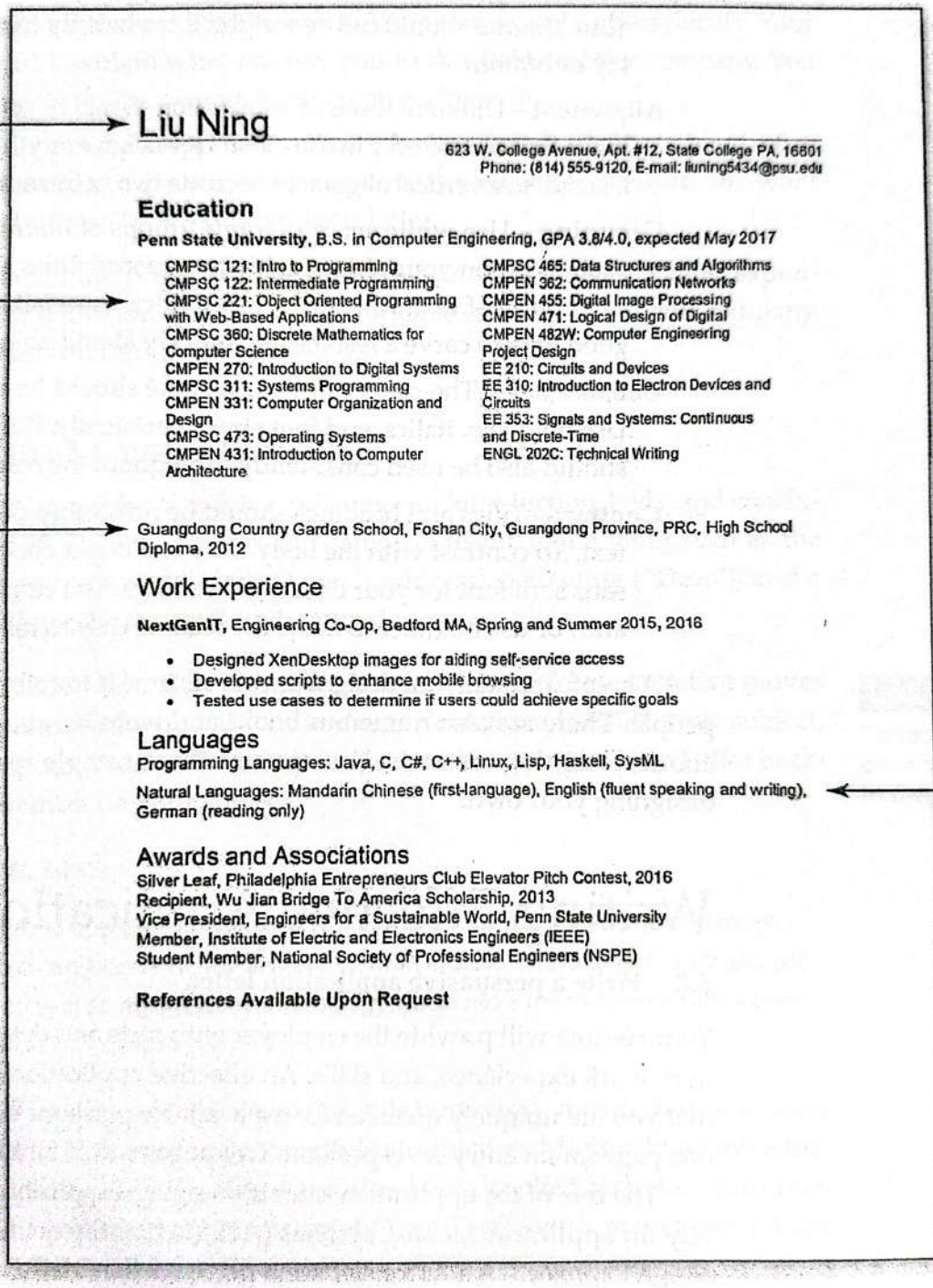
Activities and Awards

Awarded Stevenson Scholarship in Engineering, 2015
Treasurer, UMR Student Chapter, American Institute of Chemical Engineers (AIChE), 2015–Present
Member, Tau Beta Pi, 2015

Dossier with References Available at UMR Placement Services (573-555-2941)

Figure 5.3 Chronological Résumé of an International Student

Résumés for international students who want to apply for North American jobs are similar to other résumés. Language proficiency is often highlighted and sometimes a high school is mentioned.



Name is spelled as it appears on the passport.

Job-related coursework is listed here.

Résumés designed for international use will sometimes list a high school to signal a country or region of origin.

Listing natural languages and fluency will often distinguish a candidate from others.

Liu Ning

623 W. College Avenue, Apt. #12, State College PA, 16801
Phone: (814) 555-9120, E-mail: liuning5434@psu.edu

Education

Penn State University, B.S. in Computer Engineering, GPA 3.8/4.0, expected May 2017

CMPSC 121: Intro to Programming
CMPSC 122: Intermediate Programming
CMPSC 221: Object Oriented Programming with Web-Based Applications
CMPSC 360: Discrete Mathematics for Computer Science
CMPEN 270: Introduction to Digital Systems
CMPSC 311: Systems Programming
CMPEN 331: Computer Organization and Design
CMPSC 473: Operating Systems
CMPEN 431: Introduction to Computer Architecture
CMPSC 465: Data Structures and Algorithms
CMPEN 362: Communication Networks
CMPEN 455: Digital Image Processing
CMPEN 471: Logical Design of Digital
CMPEN 482W: Computer Engineering Project Design
EE 210: Circuits and Devices
EE 310: Introduction to Electron Devices and Circuits
EE 353: Signals and Systems: Continuous and Discrete-Time
ENGL 202C: Technical Writing

Guangdong Country Garden School, Foshan City, Guangdong Province, PRC, High School Diploma, 2012

Work Experience

NextGenIT, Engineering Co-Op, Bedford MA, Spring and Summer 2015, 2016

- Designed XenDesktop images for aiding self-service access
- Developed scripts to enhance mobile browsing
- Tested use cases to determine if users could achieve specific goals

Languages

Programming Languages: Java, C, C#, C++, Linux, Lisp, Haskell, SysML

Natural Languages: Mandarin Chinese (first-language), English (fluent speaking and writing), German (reading only)

Awards and Associations

Silver Leaf, Philadelphia Entrepreneurs Club Elevator Pitch Contest, 2016
Recipient, Wu Jian Bridge To America Scholarship, 2013
Vice President, Engineers for a Sustainable World, Penn State University
Member, Institute of Electric and Electronics Engineers (IEEE)
Student Member, National Society of Professional Engineers (NSPE)

References Available Upon Request

Chapter 18 offers design principles that are helpful toward creating a design for your résumé. These principles are balance, alignment, grouping, consistency, and contrast. All of these principles should be used to design your résumé.

Balance—Pay attention to the vertical and horizontal balance of the page. Your résumé should not be weighted too heavily toward the left or right, top or bottom.

Alignment—Different levels of information should be consistently indented to make the résumé easy to scan. Don't just align everything at the left margin; instead, use vertical alignment to create two or three levels in the text.

Grouping—Use white space to frame groups of information. For example, a job listed on your résumé with its responsibilities should be identifiable as a chunk of text. Sometimes using rules, especially horizontal lines, is a good way to carve a résumé into quickly identifiable sections (Figure 5.1).

Consistency—The design of your résumé should be internally consistent. Use boldface, italics, and font sizes consistently. Bullets or other symbols should also be used consistently throughout the résumé.

Contrast—Titles and headings should be noticeably different from the body text. To contrast with the body text, you might choose a different serif or sans serif font for your titles and headings. You can increase the font sizes and/or use boldface to make the résumé easy to read.

A helpful strategy for designing your résumé is to collect résumés from other people. There are also numerous books and websites available that offer ideas about designing résumés. You can use these sample résumés as models for designing your own.

Writing Effective Application Letters

5.2 Write a persuasive application letter.

Your résumé will provide the employer with facts and details about your education, work experience, and skills. An effective application letter strives to prove that you are uniquely qualified for the available position. Your letter should fit on one page for an entry-level position. Two pages would be a maximum for any job.

The role of the application letter is changing, especially as companies begin to rely on applicant tracking systems (ATS) to identify qualified people to hire or assign to projects. An ATS collects and processes the details from all submitted résumés, including college degrees, years of experience, skills, and keywords. Then, an ATS ranks resumes based on how these details are weighted by the employer.

So, letters of application can seem a bit outmoded. Are they still important? The answer is yes, but application letters are usually read when recruiters are making final decisions about whom to interview. For this reason, your application letter should do the following:

- **Show employers why you are *uniquely* qualified for the position.** If they are reading your application letter, they have already determined that you are qualified for the position. Now, show why you are the best fit for the position.
- **Show your excitement for the work.** Your résumé only offers details. Your letter should explain what attracts you to this field and this company. You need to motivate the readers to give you an interview.
- **Show your personal qualities.** Your résumé cannot illustrate your work ethic, dedication to quality, or ability to work well under pressure. You want to highlight these "soft skills" in your letter.

Your letter should prove to potential employers that your education, experience, and skills will allow you to make a valuable contribution to *their* company. Put the emphasis on *their* needs, not yours.

Content and Organization

Like any letter, an application letter will have an introduction, body, and conclusion (Figure 5.6). It will also include common features of a letter such as the header (your address and the employer's address), a greeting ("Dear"), and a closing salutation ("Sincerely") with your signature.

INTRODUCTION You should begin your letter by making up to five moves commonly found in the introduction of any document: Identify your subject, state your purpose, state your main point, stress the importance, and offer background information on your subject.

Link
For more information on writing letters, go to Chapter 6.

DEAR MS. SIMS:

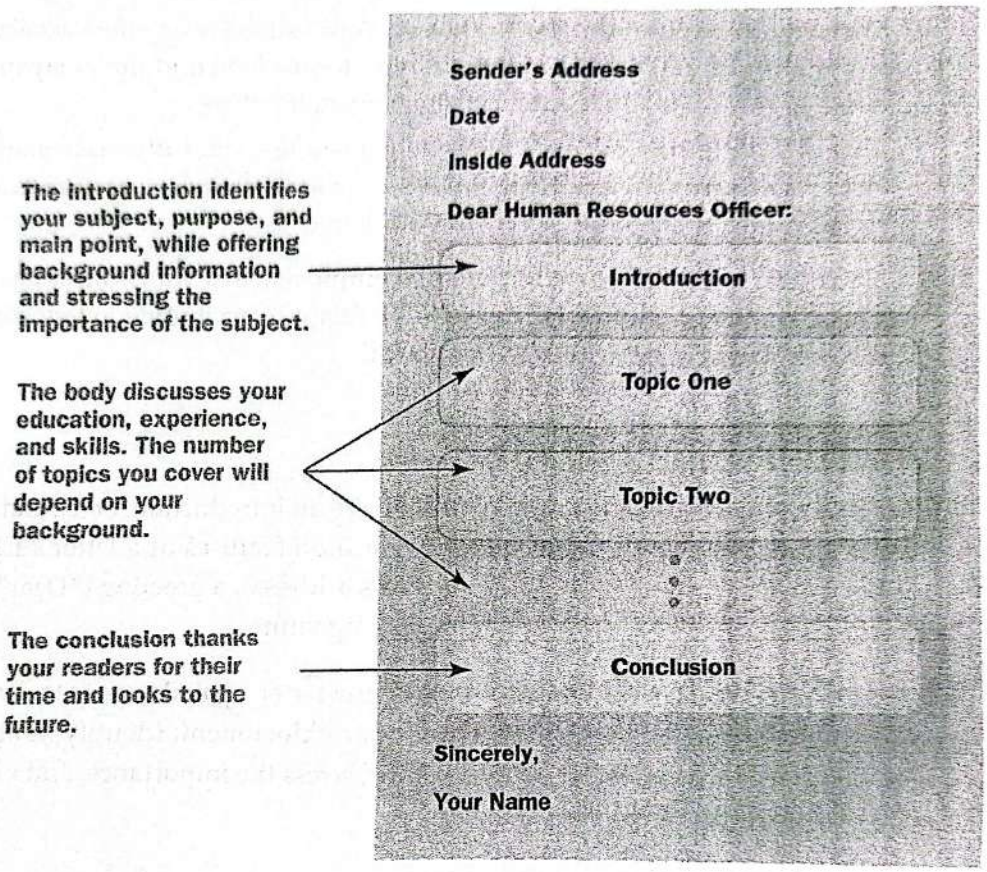
Subject and purpose → I would like to apply for the Organic Agronomist position you advertised through
Background information → Indeed.com on March 17. My experience with organic innovations in plant and soil
Main point and importance → science as well as my minor in entomology would allow me to make an immediate contribution to your company.

This brief introduction makes all five introductory moves, but you don't need to use all five moves for a successful introduction. Minimally, an introduction should identify the subject (the position being applied for), state your purpose ("I would like to apply for your job"), and state your main point ("I am uniquely qualified for your position").

BODY In the body of the letter, you should include two to three paragraphs that show how your educational background, work experience, and skills fit the employer's needs. You should organize the body of your letter to highlight your strengths. If your educational background is your best asset, put that paragraph right after the letter's introduction (Figure 5.7). If your work experience is stronger than your education, then put that information up front (Figure 5.8).

Figure 5.6 The Basic Pattern of an Application Letter

An application letter includes the common features of a letter, like the sender's address, date, inside address, introduction, body, and closing salutation.



Remember that you are making an argument, so each paragraph should start out with a claim, and the rest of the paragraph should support that claim with examples, facts, and reasoning. Here is a sample paragraph that supports a claim about an agronomist's educational background.

A clear claim → My education and research as an organic agronomist would benefit your company significantly. As a Plant and Soil Science major at Southern Illinois University, I have been studying and researching environmentally safe alternatives to pesticides. Specifically, my mentor, Professor George Roberts, and I have been working on using benevolent insects, like ladybird beetles (*Coleomegilla maculata*), to control common pests on various vegetable plants. We have also developed several varieties of organic insecticidal soaps that handle the occasional insect infestation.

Support for that claim →

In the body, you need to back up your claims with facts, examples, details, and reasoning—you need proof. You should breathe life into your letter by telling stories about yourself.

Figure 5.7 Letter of Application Emphasizing Education

A letter of application should make an argument, not simply restate items that can be found on the résumé.

834 County Line Rd.
Hollings Point, Illinois 62905

April 1, 2017

Valerie Sims, Human Resources Manager
Sunny View Organic Products
1523 Cesar Chavez Lane
Sunny View, California 95982

Dear Ms. Sims:

Opening paragraph states the subject, purpose, and main point.

I would like to apply for the Organic Agronomist position you advertised through Indeed.com on March 17. My experience with organic innovations in plant and soil science as well as my minor in entomology would allow me to make an immediate contribution to your company.

Education is discussed up front with examples.

My education and research as an organic agronomist would benefit your company significantly. As a Plant and Soil Science major at Southern Illinois University, I have been studying and researching environmentally safe alternatives to pesticides. Specifically, my mentor, Professor George Roberts, and I have been working on using benevolent insects, like ladybird beetles (*Coleomegilla maculata*), to control common pests on various vegetable plants. We have also developed several varieties of organic insecticidal soaps that handle the occasional insect infestation.

Work experience is used to show potential contributions to employer.

I also worked as an intern for Brighter Days Organic Cooperative, a group of organic farmers, who have an operation similar to Sunny View. From your website, I see that you are currently working toward certification as an organic farm. At Brighter Days, I wrote eleven agronomic plans for farmers who wanted to change to organic methods. My work experience in the organic certification process would be helpful toward earning certification for Sunny View in the shortest amount of time.

Other skills are highlighted to show unique abilities.

Finally, I would bring two other important skills to your company: a background in farming and experience with public speaking. I grew up on a farm near Hollings Point, Illinois. When my father died, my mother and I kept the farm going by learning how to operate machinery, plant the crops, and harvest. We decided to go organic in 2009, because we always suspected that my father's death was due to chemical exposure. Based on our experiences with going organic, I have given numerous public speeches and workshops to the Farm Bureau and Future Farmers of America on organic farming. My farming background and speaking skills would be an asset to your operation.

Conclusion ends on a positive note and offers contact information.

Thank you for this opportunity to apply for your opening. I look forward to hearing from you about this exciting position. I can be contacted at home (618-555-2993) or through e-mail (afranklin@unsb5.net).

Sincerely,

Anne Franklin

Figure 5.8 Letter of Application Emphasizing Work Experience

The organization of the application letter should highlight strengths. Here, work experience is being highlighted by appearing immediately after the introduction.

576 First Avenue
Rolla, Missouri 65408

March 10, 2017

Mr. Harold Brown, Human Resources Director
Farnot Plastic Solutions
4819 Renaissance Lane
Rochester, New York 14608

Dear Mr. Brown:

Opening paragraph uses background information to make personal connection.

Last week, you and I met at the University of Missouri–Rolla engineering job fair. You mentioned that Farnot Plastics might be interviewing entry-level chemical engineers this spring to work on applications of polymeric materials. If a position becomes available, I would like to apply for it. With my experience in applied rheology and polymeric materials, I would be a valuable addition to your company.

Work experience with co-op is highlighted.

My work experience includes two summers as a co-op at Vertigo Plastics, a company similar in size, products, and services to Farnot. My responsibilities included inspecting and troubleshooting the plant's machinery. I analyzed the production process and reported on the performance of the plant's operations. While at Vertigo, I learned to work with other chemical engineers in a team-focused environment.

Paragraph on education makes connections to employer's needs.

My education in chemical engineering at University of Missouri–Rolla would allow me to contribute a thorough understanding of plastics engineering to Farnot. In one of the best programs in the country, I have excelled at courses in thermodynamics, chemical process design, and chemical process materials. In addition, my work in the university's state-of-the-art chemical laboratories has prepared me to do the prototype building and vacuum forming that is a specialty of your company. I also have experience working with the CAD/CAM systems that your company uses.

Conclusion indirectly requests the interview.

The enclosed résumé highlights my other qualifications. I enjoyed speaking with you at the job fair, and I would appreciate an opportunity to talk to you again about opportunities at Farnot. If you would like more information or you would like to schedule an interview, please call me at 573-555-4391 or e-mail me at bigmondy12@umr.edu.

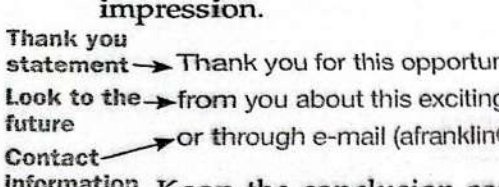
Sincerely,

J. Mondragon

James L. Mondragon

Enclosure: Résumé

CONCLUSION Your conclusion should make three moves: thank the reader, offer contact information, and look to the future. Your goal is to leave a positive impression.



Keep the conclusion concise and avoid any pleading for the position. Employers will not look favorably on someone who is begging for the job.

Style

Another way an application letter differs from a résumé is in its style. You want to adopt a style that conveys a sense of your own personality and your interest in the position available.

"YOU" ATTITUDE Put the emphasis of the letter on your readers by using the *"you" attitude*. By strategically using the words *you* and *your* in the letter, you can discuss your qualifications while putting the emphasis on the readers' needs and point of view.

Link
For more information on using the "you" style in letters, go to Chapter 6.

ACTIVE VOICE In uncomfortable situations, like writing application letters, you might be tempted to switch to passive voice. Passive voice in application letters will make you sound detached and even apathetic.

Link
For more help on using active voice, see Chapter 16.

Passive: The proposal describing the need for a new bridge over the Raccoon River was completed and presented to the Franklin City Council.

Active: My team completed the proposal for a new bridge over the Raccoon River and presented it to the Franklin City Council.

Why active voice? The active voice shows that you did something. You took action.

NON-BUREAUCRATIC TONE Avoid using business clichés to adopt an artificially formal tone. Phrases such as "per your advertisement" or "in accordance with your needs" only make you sound stuffy and pretentious.

Bureaucratic: Pursuant to your advertisement in the *Chicago Sun-Times*, I am tendering my application toward your available position in pharmaceutical research.

Nonbureaucratic: I am applying for the pharmaceutical researcher position you advertised in the *Chicago Sun-Times*.

AT A GLANCE Style in an Application Letter

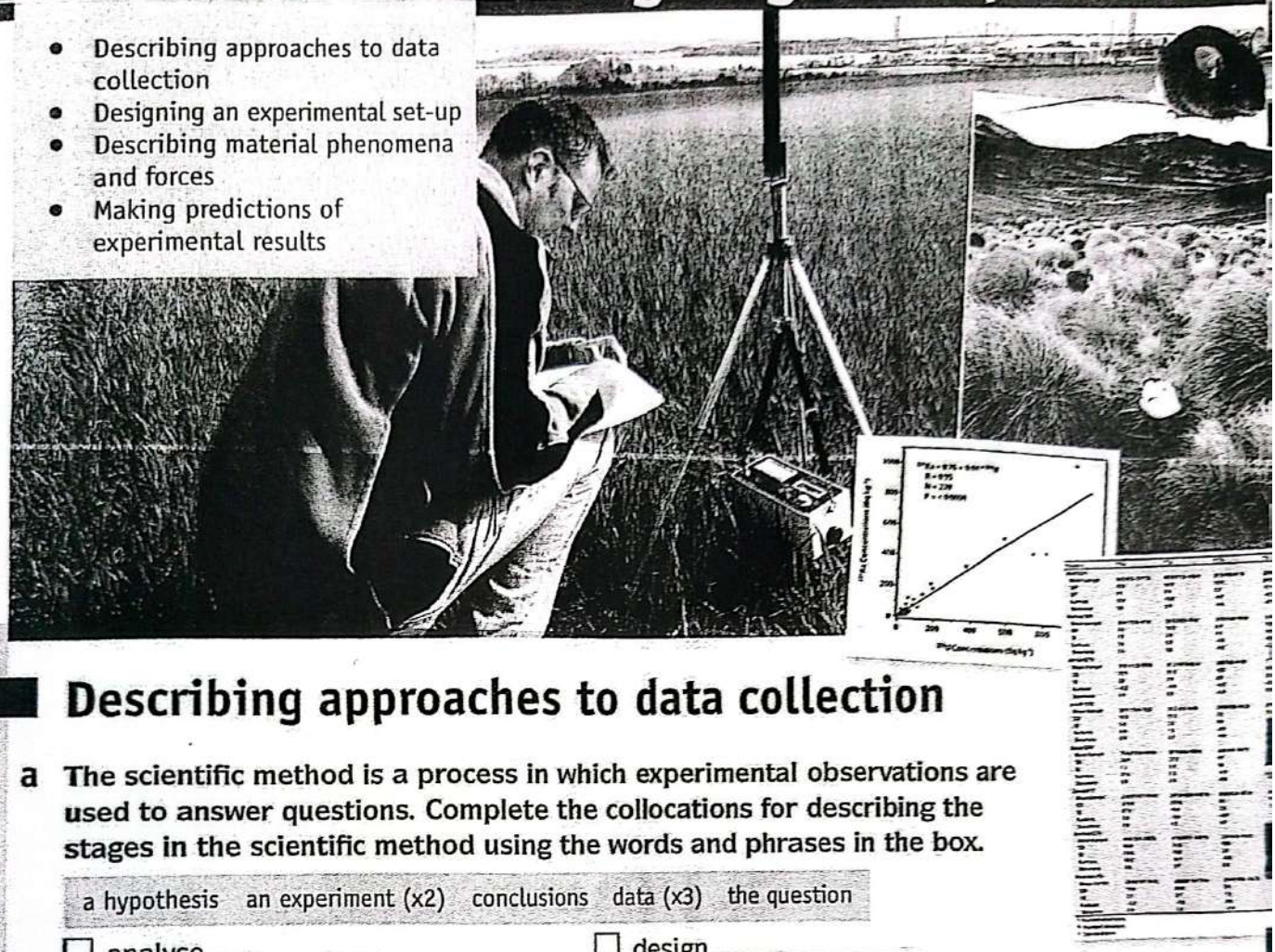
- "You" attitude—Put the emphasis on the employer.
- Active voice—Put yourself, not events, in charge.
- Non-bureaucratic tone—No one wants to hire bureaucrats, so don't sound like one.
- Themes—Point out what makes you different or attractive.

Final Term

Report Writing

UNIT 4 Designing an experiment

- Describing approaches to data collection
- Designing an experimental set-up
- Describing material phenomena and forces
- Making predictions of experimental results



Describing approaches to data collection

1 a The scientific method is a process in which experimental observations are used to answer questions. Complete the collocations for describing the stages in the scientific method using the words and phrases in the box.

a hypothesis an experiment (x2) conclusions data (x3) the question

- ☐ analyse _____
- ☐ collect _____
- ☐ conduct (or run) _____
- ☐ define _____
- ☐ design _____
- ☐ draw _____
- ☐ form _____
- ☐ interpret _____

b Number the stages (1–8) in the order you would normally do them.

c Read this extract from a student website and check your answers to Exercise 1b.

The scientific method is a process in which experimental observations are used to answer questions. Scientists use the scientific method to search for relationships between items. That is, experiments are designed so that one variable is changed and the effects of the change observed. While the exact methodologies used vary from field to field, the overall process is the same. First, the scientist must define the question – what exactly they are trying to find out. Next comes the formation of a hypothesis, which is an idea or explanation for a situation based on what is currently known. The next stage of the method is the design of an experiment which will allow this hypothesis to be tested. Usually a primary run of the experiment is conducted, and any changes to the experimental set-up made. In each experimental run, data collection takes place, followed by data analysis. Finally the data is interpreted and from this, the scientist is able to draw conclusions.

Module 9: Scientific Writing

Introduction

As a science student in higher education, you need to write scientific reports of experiments and field work. You will need to write these reports in a way that is clear enough for other scientists and readers to follow and repeat your experiments, if necessary. This means you will need to organise and structure your report in a conventional way, using conventional style, language and layout. The aim of this module is to help you learn the conventions and skills required to write a good scientific report.

Unit 1 looks at how to organise your report and timetable your work up to your submission deadline. In Units 2 and 3 you will learn conventions for writing the Method and Materials and Results sections of your report. These skills include using the passive and past tenses, and presenting and writing about tables and figures. In Unit 4, you will review conventions for writing numbers in scientific papers. Unit 5 deals with the Discussion section of the report, as well as the Bibliography, Title and Introduction. Editing and revision are essential to good writing, and you will develop these skills by doing the exercises in Unit 6. In addition, the editing check list presented in Unit 6 will be a useful guide when you come to writing full scientific reports.

Skills Map

Unit 1

Structure and schedule

Find out how to structure and timetable your report in a logical way.

Unit 2

The *Materials and Methods* section

Learn how to include appropriate content and write in a suitable style.

Unit 3

The *Results* section

Learn how to write a Results section, including how to present and describe tables and figures.

Unit 4

Writing numbers and abbreviations

Familiarise yourself with conventions for writing numbers and abbreviations in scientific reports.

Unit 5

The *Discussion, Introduction, Bibliography* and *Title* sections

Find out how to write the other sections of a report effectively.

Unit 6

Editing and revising your report

Practise editing and revising your report using an editing check list.

Destination: Scientific Writing

1 Structure and schedule

At the end of this unit you will be able to:

- structure your report and include appropriate scientific report sections;
- organise your time appropriately.

Task 1 Organising a scientific report

When you do laboratory or field work, you will be asked to write a scientific report of your experiment or investigation. Scientific papers and reports are typically divided into five sections.

1.1 Discuss the following questions in small groups. Use your answers to complete the table below.

a) What are the sections in a report?

b) What order do the sections come in?

c) What questions should be answered in each section?

Sections	Questions
1	
2	
3	
4	
5	

1.2 Feedback as a class. Discuss the reasons for your choices.

Task 2 Organising your time

When you write your report, you will not only need to write each section, but also allow time for additional research, revision and possibly discussion with your peers. You will probably have to do all this by a set deadline.

2.1 To meet deadlines, it is important to manage your time. Imagine you and your partner have to submit your reports in seven days' time. Working individually, read through the steps below and put them into a logical order. When you have finished, compare your work with your partner's.

a) Revise first draft

b) Write first draft of the *Introduction*

c) Hand in revised draft

d) Revise first draft

e) Hand in revised draft

f) Write first draft of the *Discussion*

g) Write first draft of the *Material and Methods*

h) Research background information

i) Meet with another student to discuss peer review (have a writing conference)

j) Start *Bibliography*

k) Give first draft to another student to review, using list of 'Points to check'

l) Complete practical laboratory work

m) Write first draft of the *Results* (do calculations, draw up tables, graphs, charts)

2.2 **Now that you have decided on a logical order of steps, think about how to time each step. Write your answers in the *Activity* column of the table below and give a reason for your timing, wherever possible, in the *Reason* column. Work individually.**

Time Frame	Activity	Reason
Day 0	Complete practical laboratory work	
Day 1		
Day 2		
Day 3		
Day 4		
Day 5		
Day 6		
Day 7		

Remember, you can adjust the time frame according to your own deadlines.

2.3 **Discuss your tables in small groups. You may have your own reasons for doing things in a different way from others in the group. Give reasons for your choice.**

Reflect

Think about reports you have written or read. Did they include the five sections discussed in this unit, or were the sections slightly different? How does the way a report is divided into sections help the reader?

Try to remember the different activities you included in the timetable in Task 2.2. Think about why each activity is important and how it contributes to the final report.

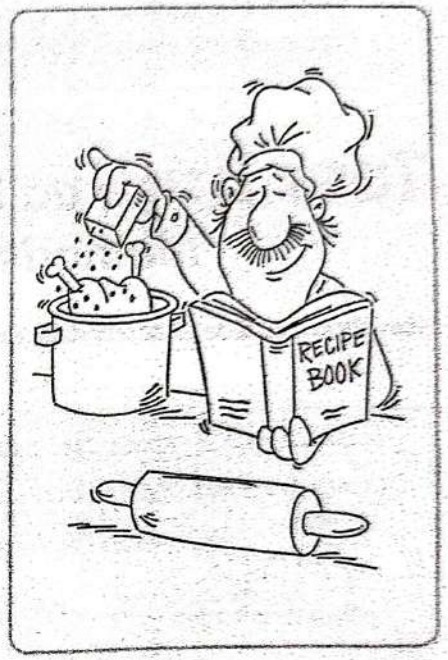
Student notes for Unit 1

Unit 2 The Materials and Methods section

At the end of this unit you will know:

- **what to include in the *Materials and Methods* section of your report;**
- **how and why to write in the passive voice.**

Task 1 What do I include?



The first section of your report is *Materials and Methods*. In this section you answer the questions: What did I do? Where? When? How?

You should describe how you did the experiment, writing in sufficient detail for another scientist to repeat your experiment.

1.1 In groups, brainstorm the following points. Choose a group secretary to write down the ideas you discuss.

a) Examples of information that should be included in a good *Materials and Methods* section, e.g., temperature, volume.

b) Examples of information that does not need to be included in a *Materials and Methods* section. (Think about your reader: what does everyone in your subject area know about the everyday equipment and techniques that you use?)

Include	Do not include

1.2 Report back to the whole class.

1.3 Which two of the following statements do you agree with?

a) Even if well-known procedures or equipment have been used, it is necessary to describe it in detail.

b) All equipment used should be listed.

c) It is important to include clear references to published protocols and methods.

d) The method should be explained using numbered points, as in a set of instructions.

Task 2 How do I write a good *Materials and Methods* section?

You should include all appropriate information in your report. It is also important to use the most appropriate style.

```
Weigh 10g air dried 2mm sieved soil into a 50ml centrifuge tube. Using an automatic
dispenser add 25ml ultra pure water. Cap the tube and place on the shaker for 15 minutes
```

pH determination

Ten g of soil was placed in a 50ml centrifuge tube. Twenty-five ml of ultra pure water was added and the tube placed on a shaker for 1 hour. The pH was then measured using a calibrated pH meter.

2.1 Complete the paragraph below, using an appropriate word from the list which follows. There are more words than gaps, so you will not need to use them all.

paragraphs sentences passive active
imperative past present future

Laboratory schedules are usually written in the _____ as a list of instructions. However, when you write your report, you must summarise what you did in full and well-developed _____. You will usually write in the _____ tense and use the _____ voice.

2.2 Discuss those questions about the advice in 2.1 as a class.

a) Why is there a difference in grammar between laboratory schedules and reports?

b) Why is information divided up into paragraphs in a report?

c) Why is the passive voice so common in the *Materials and Methods* section of a report?

2.3 As indicated above, the *Materials and Methods* section of your report, is usually written in the passive voice. The passive is used because the procedure is more important than the person who carried it out. Look at the following sentence.

I removed the skins from the onions and homogenised them in the blender.

In a scientific report this should be written:

The onions were skinned and homogenised.

In the second sentence there is no mention of the blender. Which of the following is the most likely reason?

a) It has been mentioned in a previous part of the text.

b) Only non-standard equipment should be mentioned.

c) The next sentence will state: This was carried out in a blender.

Task 3 Using the passive

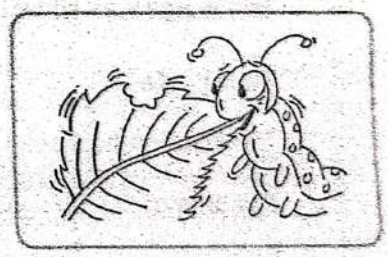
The following two sentences describe the same event.

Active:

*John conducted **the analysis.*** (***the analysis*** is the object)

Passive:

*The analysis was conducted by **John.*** (*the analysis* is the subject)



The analysis (or the methods, materials and procedures) is more important information for the reader than who conducted the analysis (John). The analysis is therefore usually made the grammatical subject of the sentence.

3.1 Complete the following rule.

The object in the active sentence, *the analysis*, becomes the _____ in the passive sentence. The subject in the active sentence, *John*, changes position in the passive sentence as it comes _____ the main verb and is introduced with _____.

3.2 Some verbs, such as *give*, have two objects.

Active:

*We gave the caterpillars **one dose** every three hours.*

We can therefore choose which object we want to make the subject. We choose the one we think is most important.

Passive:

a) *The caterpillars were given one dose every three hours.*

or

b) *One dose was given to the caterpillars every three hours.*

In the active sentence above, the two objects are _____ and _____. One or the other can become the subject in the _____ sentence.

Other common verbs that can have two objects are: *bring, send, offer, ask, pay, lend, sell.* However, you are less likely to use these in scientific reports.

3.3 Read the explanation about the use of tenses. Then complete the following sentences in the past simple.

You write the *Materials and Methods* section of a scientific report in the **past** tense. This can be the *past simple*, the *past perfect* or the *past continuous*. Each tense works the same way in the passive. It is only the verb *to be* which changes, according to the tense chosen.

Past simple

The plant was taken.	The plants _____ taken.
A pot _____ made.	The pots were made.
The image was shown.	The images _____ _____.

Past perfect

The solution had been shaken.	The solutions had been _____.
The animal _____ been fed.	The animals _____ been fed.
The mixture had been kept.	The mixtures _____.

Past continuous

A record was being made.	Records were _____.
An attachment was being fitted.	Attachments _____ being fitted.
The result was being analysed.	The results _____.

Note. In all passive sentences, *to be* is singular if the subject is singular, and plural if the subject is plural.

The box **was** being built. The boxes **were** being built.

3.4 **Look at the following draft of a student's *Materials and Methods* section and the tutor's comments on it. In small groups, discuss what changes the student should make in response to the tutor's comments. Make a note of the points you discuss in your group.**

I started my field work recordings on 12 February, 2007, and ended them on 12 March, 2007. My partner used digital camera to record the animals found on the beach and I marked the animals with quick-drying non-toxic paint. We were making recordings of environmental conditions, including the temperature, the salinity and the substrate, at the same time. I began the laboratory experiments at the same time as the field work. Each day I collected 10 animals from beach and placed them in the controlled conditions in the laboratory until experiments began.

We set up the apparatus as shown in Figure 1 and I placed 1 crab in each specimen tube. By the time an experiment started, we had acclimatised the crabs for at least 2 days. I had fed crabs daily. I had prepared their food in advance. My partner used a digital camera at the end of each experiment to record the appearance of the animals. We had printed the photographs taken at the beach for comparison. We were analysing results continuously. We analysed our results using statistical tests.

Overall, a clear description of your methods. You could improve your writing by:
- focusing the reader's attention on the method, rather than who used the methods
- paragraphing more accurately
- checking your articles (the, a, an)

3.5 **Write the final version of the section, using the tutor's comments and the notes you made to help you.**

Reflect

Look back over the unit and think about the conventions for writing scientific reports that you feel are the most important. Reflect on the reasons for your choice. For example, do you think it is important to use the passive voice, and if so, why?

Think about the sort of mistakes and omissions you might have made in the past or you feel are still likely to make. Reflect on how these mistakes might affect the way that other people regard your work.

Student notes for Unit 2

Unit 3 The Results section

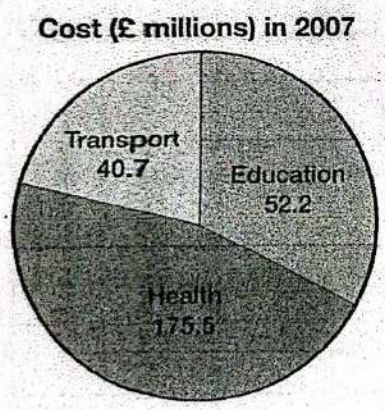
At the end of this unit you will know:
- **what to include in the *Results* section;**
- **how to present and describe tables and figures;**
- **how to write about your results.**

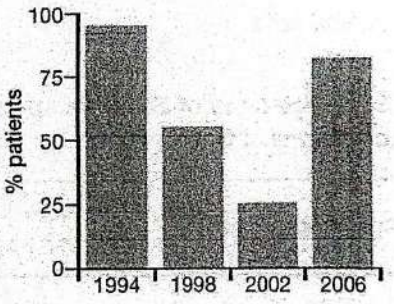
Task 1 What to include

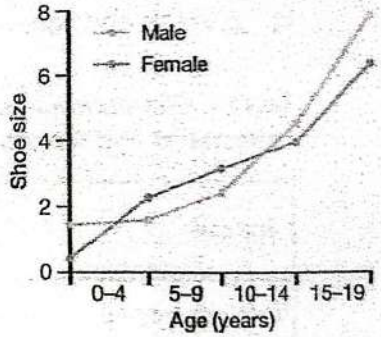
The *Results* section of your report enables you to present your data (findings or results) to show what you found and whether it matched your expectations.

This section needs to include a short paragraph or two describing and analysing trends and results, as well as any relevant tables or figures that support your findings. It may be the shortest section of your report, but it is also the most important.

1.1 Match each of the diagrams below to the correct name in the box.







Study group	N	Mean age (years)	Mean height (m)
A	17	43.2	1.77
B	24	44.9	1.69
C	22	56.2	1.64

table histogram pie chart line graph

1.2 Discuss the following questions with another student.

a) What is the difference between a table and a figure in a scientific report?

b) Should the *Results* section always include tables or figures, or can the results be expressed by text alone?

c) Should the title of the figure or table be above or below it?

d) What do you think make tables and figures easier to interpret?

Task 2 Preparing tables and graphs

If you have a large quantity of data to present, or are comparing several different things, a table can show it more clearly than a graph. Graphs and other figures, on the other hand, are a good way of illustrating and emphasising trends, particularly if they are dramatic.

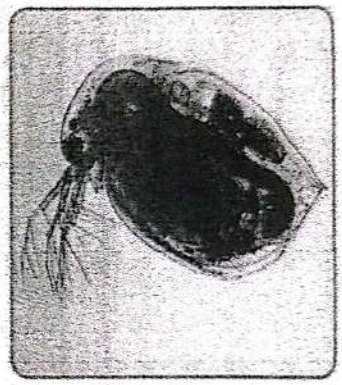
2.1 Tables 1 and 2 below present the same information but in different formats. Discuss with another student:

a) the differences between Tables 1 and 2;

b) which of the tables is better organised, easier to read and makes it easier to compare results;

c) whether it is appropriate to show this information in a graph.

2.2 Work in groups and decide what features make a good table.



Think about how to:

a) show your data to make it easy to compare significant information;

b) give units, arrange numbers, use abbreviations;

c) give table and figure numbers, and titles.

Table 1. Characteristics of three populations of Daphnia species collected at Rye Meads Pond on 3 June, 1981

Species	Average length (mm)	Average number of eggs	Average number of animals per L
Daphnia magna	5.01	15.3	112.5
D. obtusa	2.33	8.2	68.7
D. longispina	2.77	6.8	40.4

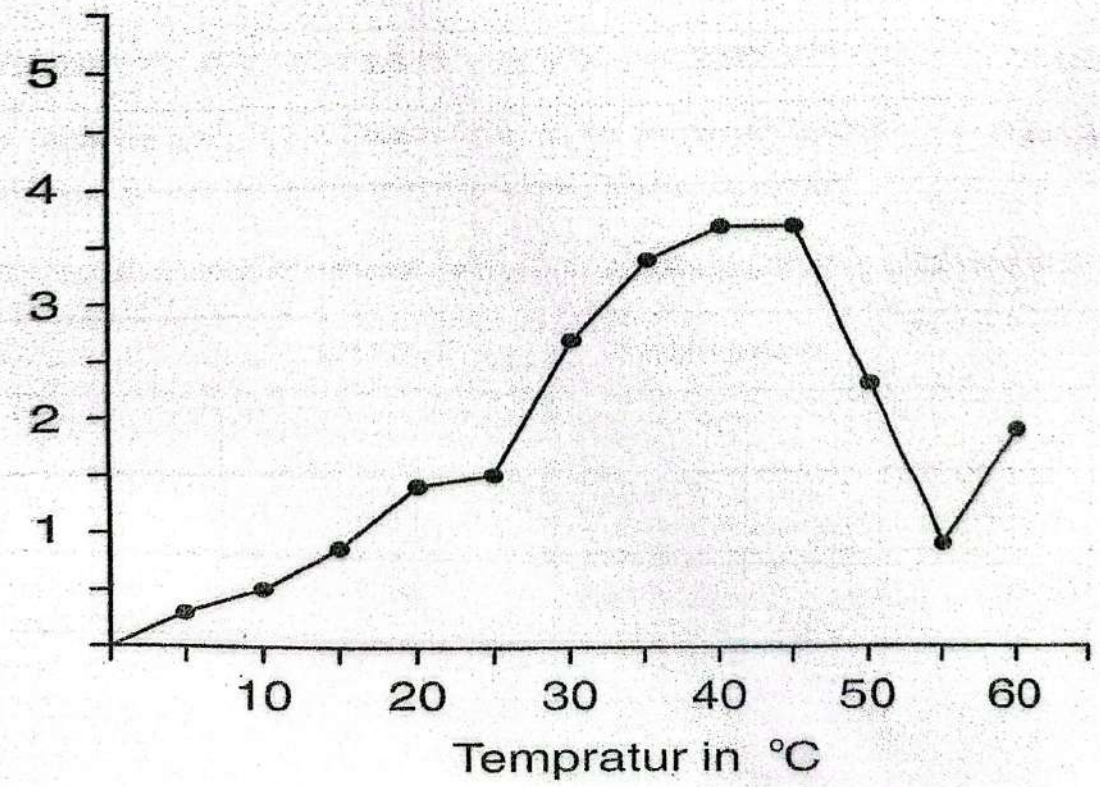
Table 2. Characteristics of three populations of Daphnia species collected at Rye Meads Pond

Species	Daphnia magna	D. obtusa	D. longispina
Av. length	5.01	2.33	2.77
Aver. no. of eggs	15.3	8.2	6.8
Av. no. of animals	112.5	68.7	40.4

2.3 The results below are taken from a student's laboratory notebook and are followed by a graph that the student made based on the data to include in a written report.

Discuss the strengths and weaknesses of the graph.

Temperature °C	Rate of reaction, mg. products per hr
0	0
5	0.3
10	0.5
15	0.9
20	1.4
25	2.0
30	2.7
35	3.3
40	3.6
45	3.6
50	2.3
55	0.9
60	0



2.4 Draw an improved version of the graph on the graph paper below.

As you prepare your graph, you should consider the following questions.

a) Which is the independent variable (the one the investigator can control or manipulate)?

b) Which is the dependent variable (the one that changes in response to the independent variable)?

c) Are the axes organised correctly? (Note: The convention is that the independent variable is usually plotted on the vertical or x- axis, and the dependent variable is plotted on the horizontal or y – axis.)

d) Are the scales appropriate?

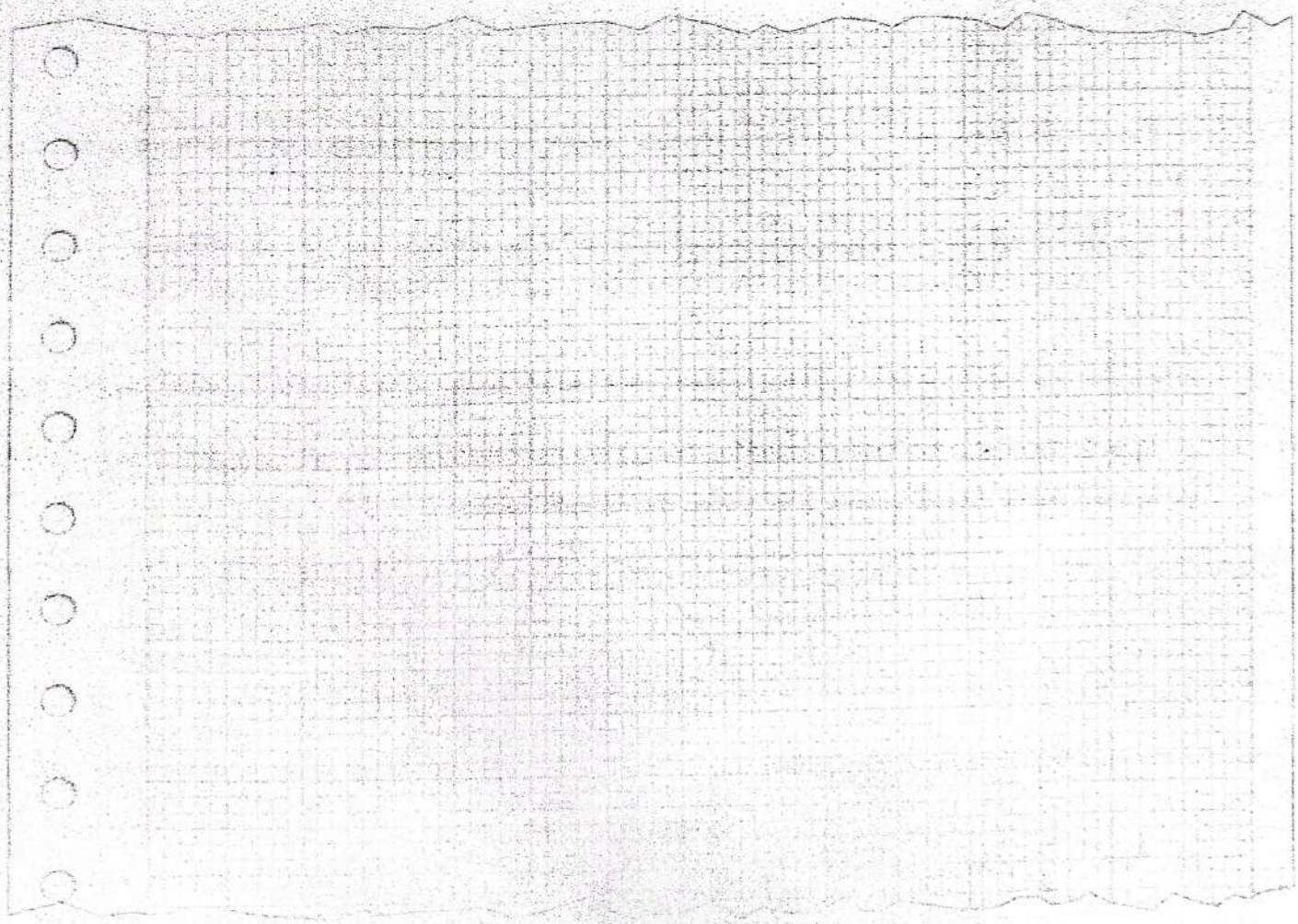
e) Are the axes labeled correctly, with units?

f) Are the data plotted accurately?

g) Are the abbreviations correct?

h) Does the graph have a figure number and a title that enables the reader to understand what the graph represents?

Note: You will usually produce graphs using a computer package (See TASK Module 7: Introduction to IT Skills).



Task 3 Writing the text of the *Results* section

After you have presented your results graphically, you must describe your findings in the rest of your *Results* section. As you are now describing what the results were, rather than what you did, you will usually use the past tense in the active voice.

Start with a sentence that states each important finding and which refers to the table or figure that supports this finding. Next, write about the specific details of the data shown in the figure.

The following example is a paragraph taken from a *Results* section of a report.

> Oxygen production varied depending on the pH of the solution (Figure 1). At pH 2, oxygen production was 3ml, whereas at pH 7 it increased to a maximum of 6ml. At pH values above 7, oxygen production decreased and was at a minimum of 1ml at pH 10.

3.1 Discuss with another student what points you would include in a description of the results presented in the graph you drew in Task 2. Then work individually and write a paragraph to describe the data.

3.2 When you and your partner have completed your paragraphs, evaluate each other's work.

a) Work in a group of 3–5. Discuss what you feel are the main features of a good description of results.

b) Write a check list giving advice for writing the *Results* section.

The Results section

Unit 5 The Discussion, Introduction, Bibliography and Title sections

At the end of this unit you will know:

- **what to include in the *Discussion* section of your report;**
- **how to cite references and write a *Bibliography*;**
- **how to write an appropriate *Title* and *Introduction*.**

The *Discussion* section is an important part of the report that follows on from the explanation of your methods and presentation of your results. After you have written the *Discussion* it will be easier to organise your *Bibliography*, *Introduction* and *Title*.

Task 1 What to include in the *Discussion*

You now have an opportunity to interpret your results and explain their significance in the *Discussion* section.

1.1 **Look at the figure below. It is a table of results for an experiment that compared the distribution of beetles in two different woodlands. Think of some questions that you might ask about the results of the experiment.**

Example:
Were there any significant differences between the two habitats?

Table 1: Mean number of beetles per quadrat for two woodlands

		Mean number per quadrat	
Scientific name	Common name	Pound Wood	West Wood
Aphodius nemoralis	A dung beetle	9.1	11.3
Curculio glandium	Acorn weevil	8.4	6.9
Coccinella septempunctata	7-spot ladybird	7.8	6.7
Adalia bipunctata	2-spot ladybird	6.3	5.9
Agonum assimilie	A ground beetle	5.1	4.9
Rhopalomesites tardyi	Holly weevil	3.9	4.2
Ampedus sanguinolentus	A click beetle	2.4	3.1
Byrrhus fasciatus	Banded pill beetle	3.1	2.5
Pogonocherus hispidulus	A longhorn beetle	2.6	1.3
Clytus arietus	Wasp beetle	3.0	3.0
Ampedus rufipennis*	A click beetle	0.8	0
Carabus intricatus*	A ground beetle	3.5	0
Dryphthorus corticalis*	A weevil	2.7	0
Gnorimus nobilis*	The noble chafer	1.7	0
Lucanus cervus*	Greater stag beetle	1.8	0

*Species of conservation importance

1.2 The paragraph below summarises what to include in the *Discussion*. Complete the gaps with words from the box.

hypothesis	suggestions	implications	conclusions	deviations

The *Discussion* section of the report will generally move from the specific (the results of your experiment) to the general (how your results fit in with other scientific findings). Normally, the discussion should do the following.

- Explain whether your results support your original _____.
- Consider any surprising data or _____ from what you expected.
- Relate your findings to previous results in the same area and derive _____ about the process you are studying.
- Look at the practical and theoretical _____ of your findings.
- Make _____ for extensions of your study.

1.3 Look at the discussion of the results in Table 1.1. on page 23.

a) Identify which different elements from Task 1.2 have been included.

b) Identify which of the following language structures and forms are used in the *Discussion*.

- the passive voice
- the past tense
- comparative structures
- modal verbs
- the imperative
- relative clauses

Discussion

The results show that populations of common beetle species were similar in both woodlands and were comparable to numbers found in previous studies. These common beetles are found in most woodland habitats and can be described as generalists. As expected, it was also found that the diversity of beetle species was higher in Pound Wood than in West Wood. Pound Wood was found to contain a surprisingly high number of beetle species which are rare in the UK and which can be described as specialists.

The results show that for beetles, Pound Wood is of more conservation interest than West Wood. It is a suitable habitat for some beetle species that are rare in Britain and are the subject of species recovery plans that aim to increase their numbers in UK habitats. Pound Wood, therefore, should take priority in management and investment for beetle conservation purposes over West Wood. However, this data says nothing of the overall biodiversity of either of the woodlands. There may be other species of conservation importance present in West Wood that have not been recorded in this study. Therefore, further work should be carried out to assess the overall biodiversity of both of these woodlands before any decisions regarding management or investment are made for either.

Task 2 Citing references and writing a bibliography

In the *Discussion* section you may compare your results with other studies. This will require you to cite references to other reports and published material.

You will also need to list all the references that you have referred to in your report in a bibliography at the end of your report. (See TASK Module 10, *Research and Referencing,* for more detailed information on citing sources and writing bibliographies). It is important to use a standard layout for this, such as the APA System.

2.1 Read the citation below and make a note of the order in which the author, publisher, date and place of publication are written.

Southwood, T.R.E. (1984). *Ecological Methods with Particular Reference to the Study of Insect Populations, 2nd ed.* New York: Chapman and Hall.

2.2 In scientific writing, many bibliography references will be to articles in scientific journals. Look at the two citations below and answer the following questions.

a) What do the numbers 13,2025-2036 in the second citation refer to?

b) What abbreviations are used in the second citation?

c) What is the main difference between the two citations?

Wallace, M.J., Newton, P.M., Oyasu, M., McMahon, T., Chou, W.H., Connolly, J., and Messing, R.O. (2006). *Acute Functional Tolerance to Ethanol Mediated by Protein Kinase C Varepsilon.* Neuropsychopharmacology, Published online.

Yamada, K., Fukaya, M., Shimizu, H., Sakimura, K., and Watanabe, M. (2001). *NMDA receptor subunits Glurepsilon 1, GluRepsilon 3 and GluRzeta 1 are encircled at the mossy fibre-granule cell synapse in the adult mouse cerebellum.* Eur. J. Neurosci. *13, 2025-2036.*

2.3 Compare your answers with another student and discuss the bibliography conventions.

It is advisable to start your bibliography with full details and in the correct format as soon as you start reading references. Then you can add references to your list as you go along. In this way you will find that you avoid omissions and errors, and will save yourself time.

To help you do this, always keep a detailed record of your references; for example, make notes when reading in the library without a computer.

When you have finished the report, give your Bibliography a final check, making sure it is complete and presented in the correct format.

Task 3 What to include in the *Introduction*

Now that you have written the *Materials and Methods*, *Results* and *Discussion* sections, you are in a position to write an *Introduction* to your report.

3.1 Look at the questions and decide which ones might be addressed in the *Introduction*. Discuss your ideas in groups.

a) When did you do your experiment?

b) What was the background of your experiment?

c) What was the aim of your experiment?

d) Were there any unexpected results?

3.2 Look at the following example of an *Introduction* and put the sentences in the correct order.

Beetles in Woodland Habitats

Introduction

a) Many of these species are the subject of species recovery plans designed to manage suitable habitat and increase their numbers. Organisations with responsibilities for areas of woodlands are often lacking in sufficient resources to protect the entire woodland habitat under their jurisdiction.

b) Woodland habitats have been in decline throughout Britain for centuries. This decline has been most notable since the Industrial Revolution and the mechanization of farming practices.

c) Therefore, they have to prioritise woodlands that are in need of immediate protection.

d) The organisms under most threat are the plants and the insects. These organisms tend to have low dispersal rates and are slow to colonise new habitats. Many species of plants and invertebrates are now threatened with extinction in Britain due to a loss of habitat.

e) With the decline in woodland, many organisms are under threat from a loss of habitat.

f) This study assesses the importance of two woodland habitats to beetle conservation.

Task 4 What makes a good *Title*?

The *Title* gives the reader a concise and informative description of the focus of your report. It summarises the information contained in the *Introduction* and *Results* sections. You may use a 'working title' during the writing stages, but you should revise it when your report is complete.

The title should give the reader a complete description of the study and include important keywords and phrases.

4.1 **Look at the following titles. Work in groups and decide which one is better and why.**

Title 1: *Determination of metabolic rate*

Title 2: *The effect of temperature on oxygen consumption in mice*

4.2 **Discuss your findings as a class.**

4.3 **Work with another student and compare the following two pairs of titles. Think about the different information each pair gives the reader. Come to a clear conclusion about the importance of an appropriate title.**

Title 1: *Species composition of summer phytoplankton in Lake Windermere, Great Britain*

Title 2: *Sampling plankton in a lake*

Title 1: *Effects of pollutants on Daphnia*

Title 2: *Morphological and ultrastructural effects of sublethal cadmium poisoning on Daphnia species*

Reflect

You have now done work on all aspects of a piece of scientific writing. Think about what you have learnt in Units 1 to 5 and how the information in each unit ties in with the others.

Student notes for Unit 5

6 Editing and revising your report

At the end of this unit you will:
- be more aware of how to check your work for grammatical and vocabulary mistakes;
- be more aware of how to edit your work to ensure that you use full sentences which are clear and concise.

Task 1 What do I check for?

After you have written your report, it is important to check for mistakes and errors. You will get into the habit of editing more efficiently if you are aware of the key areas in which you tend to make mistakes.

'The high temperature Sp. effected the results.'

1.1 Working in groups, look at the questions below and discuss which problem areas you feel are the most important.

- Is the use of vocabulary appropriate?
- Is the general vocabulary correctly spelt?
- How about the specialist vocabulary? Do you need to look any of it up?
- Does the numbering follow the correct conventions?
- Does your punctuation help the reader?
- Is writing clear and concise?
- Can you spot any errors in grammar?

1.2 Compile a check list of problem areas you should check to revise and edit a report.

1
2
3
4
5
6
7
8

Task 2 Use of tenses

In this module, you have looked at the academic conventions for use of different tenses and aspects in scientific reports. It is important to remember to use these correctly and consistently in your work.

2.1 The table below shows sections and topics that might be included in a scientific report. For each one, choose whether the past or present tense should be used in the example sentence.

Section of report	Example sentence
a) *The Materials and Methods* section	The apparatus ***is /was*** set up as shown.
b) Referring to a table or graph	Table 1 ***presents / presented*** the results from sites 3 and 4.
c) Stating (quoting) the findings of published work	Cadmium ***is / was*** a highly toxic metal to freshwater fish (Ball, 1999).
d) The *Results* section	Oxygen production ***varies / varied*** depending on the pH of the solution.
e) Referring to someone else's work	Smith (2002) ***finds / found*** that ...
f) To make a general statement	Respiration ***is / was*** a complex series of chemical reactions that ***results / resulted*** in the release of energy from food.

2.2 Identify any sentence(s) which use the passive voice. Then discuss why you think it should be used.

Task 3 Common mistakes with vocabulary

The following exercise identifies some words that are commonly confused with each other in scientific writing.

3.1 Look at the following pairs of sentences. Choose the correct word to complete each one.

a) affect / effect

1) Temperature strongly _____ the rate of reaction.

2) The study investigated the _____ of temperature on rate of reaction.

b) continual / continuous

1) The tank was provided with a _____ supply of nitrogen.

2) Impurities can be eliminated by _____ heating, cooling and reheating.

c) site / cite

1) Smith's study several previous incidents.

2) The chosen for the experiment was a nearby pond.

d) their / there

1) is more than one way to do this.

2) The crabs were fed daily and food supply was adjusted gradually.

3.2 Now write sentences of your own for each of the pairs of words below. Use a standard dictionary to check the meaning and spelling of terms.

a) fewer / less

b) breath / breathe

c) rise / raise

d) consecutive / concurrent

3.3 Compare your ideas with another student.

Task 4 Plurals

Scientific words often have irregular plurals, particularly if they come from Greek or Latin terms. You will need to notice and remember common patterns, such as those on the next page.

You should also check that you are consistent in your use of plural nouns and that plural subjects agree with the verb that follows them.

4.1 Complete the following table of singulars and plurals of common scientific terms.

Singular	Plural
analysis	
	bacteria
	criteria
datum	
formula	
hypothesis	
medium	
ratio	
	phenomena

4.2 Complete the following sentences.

a) Greek- or Latin-based singular nouns that end in *–um* generally form the plural by changing _____.

b) Greek- or Latin-based singular nouns that end in *–is* generally form the plural by _____.

4.3 Decide whether the subjects and verbs agree in the sentences below. Correct them if necessary. (Note: Two sentences are correct and two need to be changed.)

a) This data is supported by evidence from other studies.

b) 10 drops of hydrochloric acid were added to each sample.

c) The period of immersion for crabs at different times of the tidal cycle are presented in Table 1.

d) One source of error in these experiments are the inaccuracy in recording light intensities.

Task 5 Be clear and concise

When writing reports, it is important to consider your audience. Unless you take great care to write clearly, it is easy to confuse the reader, particularly if you are describing a complex experiment or set of results.

One way to ensure clarity is to make sure that you do not use more words than necessary.

5.1 Work with another student or in small groups. Discuss how you would revise the following sentences to eliminate unnecessary words.

Example:
In the experiment, the test animals were subject to analysis for investigation of their gut contents.

The gut contents of the test animals were analysed.

a) One of the environmental conditions to which the zooplankton were shown to be affected by was pH.

b) The experiments alone are insufficient to tell what the optimum conditions are.

c) Ten test tubes were labeled with the following concentrations of sodium chloride and 50 ml of those solutions were then prepared and placed in the test tubes: 0%. 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%.

Task 6 Write in complete sentences

Another reason why reports are sometimes difficult to read is that they are not written using complete, well-formed sentences.

6.1 With another student, revise a, b and c to make complete well-formed sentences. When you have written your paragraphs, compare them with other pairs in the group.

Example:

In accordance with the Law of Limiting Factors, rate of photosynthesis is affected by light intensity, temperature and carbon dioxide concentration. Resulting in maximum rates in optimum conditions.

In accordance with the Law of Limiting Factors, rate of photosynthesis is affected by light intensity, temperature and carbon dioxide concentration and maximum rates occur in optimum conditions.

a) In the third set of experiments, citric acid concentration was doubled and at each temperature three sets of readings.

b) Enzymes are denatured at high temperatures. Because molecular conformation is altered.

c) The reaction occurred at its maximum; copper was absent.

Reflect

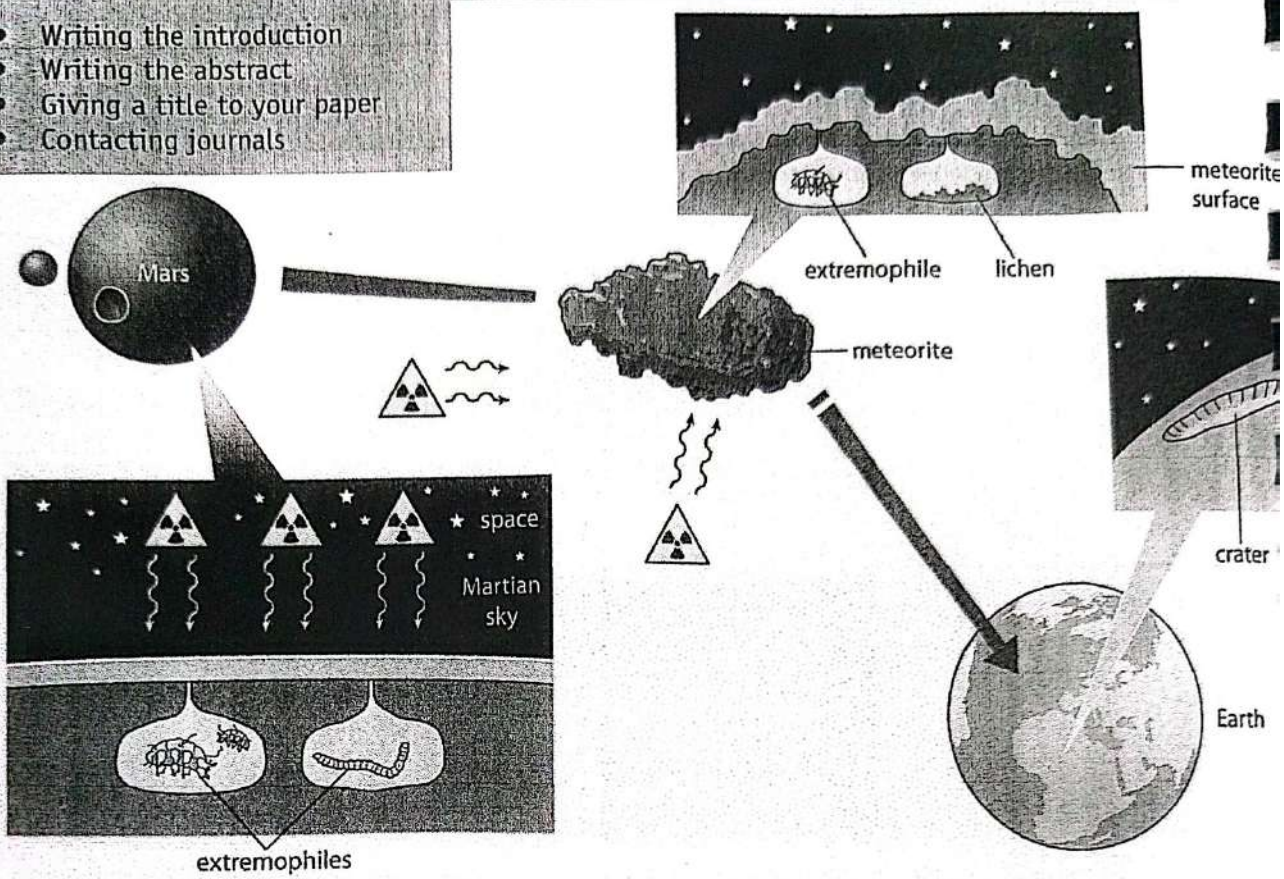
Collect samples of reports that have been submitted by fellow students. Practise noticing where things could be improved.

As you carry this out, develop your own check list and procedure for editing.

Student notes for Unit 6

UNIT 9 Writing up research 4: introduction and abstract

- Writing the introduction
- Writing the abstract
- Giving a title to your paper
- Contacting journals



Writing the introduction

1 a **Mya is doing research into the panspermia hypothesis as part of a Master's degree in astrobiology. In pairs, use the diagram and the words in the box below to help you answer questions 1–3. Then compare your ideas with the Answer key on page 114.**

1 What do you think the panspermia hypothesis is?
2 How credible do you think the hypothesis is?
3 What kind of evidence would support this hypothesis?

deep space lichen seeds of life extremophile meteorite UV radiation
harsh conditions protective layer vehicle

b **Mya has been investigating whether it is possible for bacteria and microorganisms to survive in an environment as harsh as the surface of Mars. He has been advised to organise the text of his introduction around five key questions. Match the beginnings to the endings of the questions.**

What was I	approach the problem?
Why was it	expect to know after doing the research?
What was already	important?
What did I	investigating?
How did I	known about the subject of my research?

c **Read five extracts from the introduction to Mya's paper, ignoring the highlighted words for now. Which question from Exercise 1b is each extract answering? Write the questions above the extracts.**

1 _____

Such an extreme environment was thought to be uninhabitable, but microbial ecology studies reported the presence of microorganisms (Amaral-Zettler *et al.*, 2002). Could the surface composition of Mars protect life against radiation?

2 _____

A number of studies have investigated different extreme Martian surface conditions on terrestrial microorganisms. Nicholson and Schuerger (2005) reported that the spores of *Bacillus subtilis* were able to survive for 19 days under Mars atmospheric pressure and composition. Saffary *et al.* (2002), however, found that survival decreased due to ...

3 _____

Potential habitability in the subsurface would increase if the overlaying material did play a protective role.

4 _____

For many years now, scientists have speculated about the possibility of life on Mars (Klein *et al.*, 1976; McKay, 1997). The discovery of liquid water on Mars would increase its habitability ...

5 _____

We report here on our studies of protection by Río Tinto Basin iron oxides and hydroxides on two microorganisms, *Acidithiobacillus ferrooxidans* and *Deinococcus radiodurans*, under simulated Mars surface conditions.

a **A well-written introduction usually presents general information about the topic first before specific information about the research. What do you think is the best order for the extracts in Exercise 1c?**

b **Answer the following questions about the extracts in Exercise 1c.**

1 Write down the highlighted words and phrases which describe:

a a hypothesis / hypothetical situation

_____

b current research or knowledge on a particular topic

_____

c general statements about past research

_____

d the results or conclusions taken from specific past research

_____

2 What form or tense are the words and phrases in a–d above?

3 a **Read an extract from the introduction of a paper about the ability of lichens and microbes to survive in deep space. Ignoring the gaps, how many of the questions in Exercise 1b can you answer?**

Recent advances in space technology (1) _____ (provide) the possibility of studying the survival of different microorganisms in the harsh environment of space (Demets *et al.*, 2005; Baglioni *et al.*, 2007). So far, lichens (2) _____ (be) the only organisms able to survive exposure to such extreme conditions (Sancho *et al.*, 2007; de los Ríos *et al.*, 2010).

It is believed that, if sufficiently protected by meteorite-like material, microorganisms may also survive the journey through space. However, Brandstätter *et al.* (2008) (3) _____ (report) that microorganisms embedded in 2 cm thick rocks on the outer surface of a re-entry capsule, simulating the entry of a meteorite, (4) _____ (not survive).

The aim of this work (5) _____ (be) to obtain further information on the resistance of rock-colonising microbial communities and lichens to outer space conditions, during the Biopan-6 flight of ESA on board a Russian Foton satellite.

b **Complete the extract by writing the verbs in brackets in the correct tense.**

4 a **Read two sentences which present the same information in different ways. In pairs, try to decide what the main difference is between the two sentences. Then compare your ideas with the Answer key on page 115.**

a Amaral-Zettler *et al.* (2002) reported the presence of microorganisms.
b Microbial ecology studies detected the presence of microorganisms (Amaral-Zettler *et al.*, 2002).

b **Complete the following sentences with phrases a–d below.**

1 We usually use an author-prominent citation when _____ and when _____.
2 An information-prominent citation is usually used when _____ and when _____.

a dealing with ideas we wish to explore in a paper
b dealing with information which is not controversial
c comparing ideas from a variety of sources
d supporting a particular point

c **A number of different reporting verbs can be used in author-prominent citations. Complete the definitions using the reporting verbs in the box.**

conclude demonstrate discover hypothesise observe prove suggest

(1) _____ or (2) _____ : to use an experiment to show that something is true
(3) _____ : to carefully watch the way something happens, then record it
(4) _____ and (5) _____ : to give a possible explanation for something which has not been proved
(6) _____ : to decide something after thinking about it carefully
(7) _____ : to find or learn information, especially something new

5 a **In pairs, take turns to ask and answer the questions in Exercise 1b about a piece of research you know well. Make a note of your answers.**

b **Use the notes you made in Exercise 5a to write a short introduction to your research. Remember to include author-prominent and information-prominent references in your introduction.**

Writing the abstract

6 a **In pairs, discuss the following questions.**

1 What is the purpose of an abstract?
2 How can an abstract help a researcher choose which papers to read?
3 What information does the abstract usually include?
4 Why do some people think a good abstract is even more important in the internet age than it was before?

b **An abstract usually contains one or two key sentences from each section of a paper. Read the following extracts from Mya's draft abstract. Match a section (1–4) to an extract (A–D).**

1 Introduction _____ 3 Results _____
2 Method _____ 4 Discussion _____

A

With the aim of evaluating this possibility two microorganisms, *Acidithiobacillus ferrooxidans*, an acidophile, and *Deinococcus radiodurans*, a radiation-resistant microorganism, were exposed to simulated Mars conditions; that is, 95% CO_2, 2.7% N_2, 1.6% Ar and 0.6% H_2O with a pressure of 7 mbars. Temperature was set at 150 K and ultraviolet radiation was in the wavelength range of 200–400 nmat. Exposure was for different times under the protection of 2 and 5 mm layers of oxidised iron minerals. Survival was evaluated by growing the organisms on fresh media.

B

The resistance of organisms to extreme conditions like the conditions which exist on the surface of Mars under the protection of a thin material layer increases the possibility that life could exist on Mars.

C

Here we report that both the 2 and 5 mm thick layers provided enough protection against radiation and Mars environmental conditions for the bacteria to survive (Figs. 2 & 3).

D

Current surface conditions on Mars are extremely challenging for life. However, Nicholson and Schuerger (2005) reported that *Bacillus subtilis* was able to survive for 19 days under Mars atmospheric pressure and composition. The question is whether there are any features on Mars that could provide protection against the surface conditions. One possibility is that the surface material plays a protective role due to the fact that it is composed of iron oxides and hydroxides.

c **In pairs, decide on the best order for the extracts (A–D) in the abstract. Give reasons for your answer.**

a ▶ 9.1 **Svenja, Mya's supervisor, is commenting on the draft abstract in Exercise 6b. Listen to part of the conversation and say which section (A–D) Svenja does *not* comment on.**

How to Cite References: IEEE Documentation Style

IEEE citation style is used primarily for electronics, engineering, telecommunications, computer science, and information technology reports. The three main parts of a reference are as follows:

- Author's name listed as first initial of first name, then full last.
- Title of article, patent, conference paper, etc., in quotation marks.
- Title of journal or book in italics.

This system allows the reader to identify the information source at a glance. All punctuation, dates, and page numbers depend on the type of reference cited, so follow the examples with care. Please note this style guide is wide ranging, but not all sources are identified. Further research may be required.

Citation Within The Text

The first step of the reference citation process is within the report itself. Each citation must be noted within the text through use of simple sequential numbers. A number enclosed in square brackets, placed in the text of the report, indicates the specific reference. Citations are numbered in the order in which they appear. Once a source has been cited, the same number is used in all subsequent references in the report. No distinction is made between electronic and print sources, except in the citation reference details.

Each reference number should be enclosed in square brackets on the same line as the text, before any punctuation, with a space before the bracket.

Examples

". . .end of the line for my research [13]."

"The theory was first put forward in 1987 [1]."

"Scholtz [2] has argued. . . ."

"For example, see [7]."

"Several recent studies [3, 4, 15, 22] have suggested that. . . ."

Note: Authors and dates do not have to be written out after the first reference; use the bracketed number. Also, it is not necessary to write "in reference [2]." Just write "in [2]."

The preferred method to cite more than one source at a time is to list each reference in its own brackets, then separate with a comma or dash:

[1], [3], [5]
[1] – [5]

Reference Lists

To finish citing sources, a numbered list of references must be provided at the end of the paper. The list is comprised of the sequential enumerated citations, with details, beginning with [1], and is not alphabetical.

Page Format

- Place references flush left
- Single-space entries, double-space between.
- Place number of entry at left margin, enclose in brackets.
- Indent text of entries.

The following examples demonstrate the format for a variety of electronic and print sources. These citations are ones in widest use. Not everything is listed.

Electronic Documents

E-books

[1] L. Bass, P. Clements, and R. Kazman, *Software Architecture in Practice*, 2nd ed. Reading, MA: Addison Wesley, 2003. [E-book] Available: Safari e-book.

Article in Online Encyclopedia

[2] D. Ince, "Acoustic coupler," in *A Dictionary of the Internet*. Oxford University Press, [online document], 2001. Available: Oxford Reference Online, http://www.oxfordreference.com [Accessed: May 24, 2007].

Journal Article Abstract (accessed from online database)

[1] M. T. Kimour and D. Meslati, "Deriving objects from use cases in real-time embedded systems," *Information and Software Technology*, vol. 47, no. 8, p. 533, June 2005. [Abstract]. Available: ProQuest, http://www.umi.com/proquest/. [Accessed November 12, 2007].

Journal Article in Scholarly Journal (published free of charge on the Internet)

[2] A. Altun, "Understanding hypertext in the context of reading on the web: Language learners' experience," *Current Issues in Education*, vol. 6, no. 12, July, 2005. [Online serial]. Available: http://cie.ed.asu.edu/volume6/number12/. [Accessed Dec. 2, 2007].

Newspaper Article from the Internet

[3] C. Wilson-Clark, "Computers ranked as key literacy," *The Atlanta Journal Constitution*, para. 3, March 29, 2007. [Online], Available: http://www.thewest.com.au. [Accessed Sept. 18, 2007].

Internet Documents

Professional Internet Site

[1] European Telecommunications Standards Institute, "Digital Video Broadcasting (DVB): Implementation guide for DVB terrestrial services; transmission aspects," *European Telecommunications Standards Institute*, ETSI-TR-101, 2007. [Online]. Available: http://www.etsi.org. [Accessed: Nov. 12, 2007].

General Internet Site

[2] J. Geralds, "Sega Ends Production of Dreamcast," *vnunet.com*, para. 2, Jan. 31, 2007. [Online]. Available: http://nli.vnunet.com/news/1116995. [Accessed Sept. 12, 2007].

Personal Internet Site

[3] G. Sussman, "Home Page-Dr. Gerald Sussman," July, 2002. [Online]. Available: http://www.comm.edu.faculty/sussman/sussmanpage.htm. [Accessed Nov. 14, 2007].

Email

[4] J. Aston. "RE: new location, okay?" Personal email (July 3, 2007).

Internet Newsgroup

[5] G. G. Gavin, "Climbing and limb torsion #3387," USENET: sci.climb.torsion, August 19, 2007. [Accessed December 4, 2007].

Microform

[6] W. D. Scott, *Information Technology in the US*. [Microform]. W. D. Scott & Co., Canberra: Department of Science and Technology, 2004.

Computer Game

[7] The Hobbit: *The prelude to the Lord of the Rings*. [CD-ROM]. United Kingdom: Vivendi Universal Games, 2003.

Software

[8] Thomson ISI, *Endnote 7*. [CD-ROM]. Berkeley, CA: ISI ResearchSoft, 2006.

Lecture

[1] S. Bhanndahar. ECE 4321. Class Lecture, Topic: "Bluetooth can't help you." School of Electrical and Computer Engineering, Georgia Institute of Technology, Atlanta, GA, Jan. 9, 2008.

Print Documents

Books

Single Author

[1] W. K. Chen, *Linear Networks and Systems*. Belmont, CA: Wadsworth Press, 2003.

Edited Book

[2] J. L. Spudich and B. H. Satir, Eds., *Sensory Receptors and Signal Transduction.* New York: Wiley-Liss, 2001.

Selection in an Edited Book

[3] E. D. Lipson and B. D. Horwitz, "Photosensory reception and transduction," in *Sensory Receptors and Signal Transduction*, J. L. Spudich and B. H. Satir, Eds. New York: Wiley-Liss, 2001, pp-1-64.

Three or More Authors

[4] R. Hayes, G. Pisano, and S. Wheelwright, *Operations, Strategy, and Technical Knowledge*. Hoboken, NJ: Wiley, 2007.

Book by an Institutional or Organizational Author

[5] Council of Biology Editors, *Scientific Style and Format: The CBE Manual for Authors, Editors, and Publishers*, 6th ed., Chicago: Cambridge University Press, 2006.

Manual

[6] Bell Telephone Laboratories Technical Staff, *Transmission System for Communication*, Bell Telephone Lab, 2005.

Application Note

[7] Hewlett-Packard, Appl. Note 935, pp.25-29.

Note: Titles of unpublished works are not italicized or capitalized. Capitalize only the first word.

Technical Report

[8] K. E. Elliott and C. M. Greene, "A local adaptive protocol," Argonne National Laboratory, Argonne, France, Tech. Report. 916-1010-BB, 7 Apr. 2007.

Patent/Standard

[9] K. Kimura and A. Lipeles, "Fuzzy controller component," U. S. Patent 14, 860,040, 14 Dec., 2006.

Data Sheet

[10] Texas Instruments, "High speed CMOS logic analog multiplexers/demultiplexers," 74HC4051 datasheet, Nov. 1997 [Revised Sept. 2002].

Government Publication

[11] National Aeronautics and Space Administration, *NASA Pocket Statistics*. Washington, DC: Office of Headquarters Operations, 2007.

Paper Published in Conference Proceedings

[12] J. Smith, R. Jones, and K. Trello, "Adaptive filtering in data communications with self improved error reference," In Proc. IEEE International Conference on Wireless Communications '04, 2004, pp. 65-68.

Papers Presented at Conferences (unpublished)

[13] H. A. Nimr, "Defuzzification of the outputs of fuzzy controllers," presented at 5th International Conference on Fuzzy Systems, Cairo, Egypt, 2006.

Thesis or Dissertation (unpublished)

[14] H. Zhang, "Delay-insensitive networks," M. S. thesis, University of Chicago, Chicago, IL, 2007.

Article in Encyclopedia, Signed

[15] O. Singh, "Computer graphics," in *McGraw-Hill Encyclopedia of Science and Technology*, New York: McGraw-Hill, 2007, pp. 279-291.

Journal Articles

Article in Journal (paginated by annual volume)

[8] K. A. Nelson, R. J. Davis, D. R. Lutz, and W. Smith, "Optical generation of tunable ultrasonic waves," *Journal of Applied Physics*, vol. 53, no. 2, Feb., pp. 1144-1149, 2002.

Article in Professional Journal (paginated by issue)

[9] J. Attapangittya, "Social studies in gibberish," *Quarterly Review of Doublespeak*, vol. 20, no. 1, pp. 9-10, 2003.

Article in Monthly or Bimonthly Periodical

[10] J. Fallows, "Networking technology," *Atlantic Monthly*, Jul., pp. 34-36, 2007.

Article in Daily, Weekly, or Biweekly Newspaper or Magazine

[11] B. Metcalfe, "The numbers show how slowly the Internet runs today," *Infoworld*, 30 Sep., p. 34, 2006.

IEEE Reference Writing Practice

Information needed for a Journal Article with a DOI (Digital Object Identifier)
Author: Heather L. O'Brien and Elaine G. Toms
Date: April 2008
Title: What is user engagement? A conceptual framework for defining user engagement with technology
Journal: Journal of the American Society for Information Science and Technology (JASIST)
Specifics: Volume 59, Issue 6, pages 938-955
DOI: 10.1002/asi.20801
Note: The DOI is a unique number used to identify a specific article.

Information needed for a Journal Article (multiple formats online, URL to a directory)
Author: E.F. Vasechkina and V.D. Yarin
Date: 2001
Title: Evolving polynomial neural network by means of genetic algorithm: some application examples
Journal: Complexity International
Specifics: Volume 09
URL: http://www.csu.edu.au/ci/vol09/vasech01/
Note: The URL goes to a directory where the format is selected.

Information needed for a Book (by one or more authors)
Author: Trevor Hastie, Robert Tibshirani, and Jerome H. Friedman
Date: 2001
Title: The elements of statistical learning: data mining, inference, and prediction
Place where published: New York, NY
Publisher: Springer-Verlag

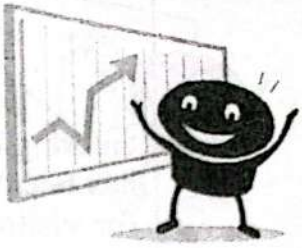
Information needed for an Authored Chapter in a Book (by one or more editors)
Author: M.T. Maybury
Date: 2001
Title of Chapter: Intelligent user interfaces for all
Title of Book: User interfaces for all: concepts, methods and tools
Editor: Constantine Stephanidis
Place Book Published: Mahwah, New Jersey
Publisher: Lawrence Erlbaum Associates
Pages: 65-80

Information needed for a Webpage (html file, URL significant)
Author: Jakob Nielsen
Date: 1994
Title: Heuristics for User Interface Design
Page: Ten Usability Heuristics
URL: http://www.useit.com/papers/heuristic/heuristic_list.html

Describing Graphs & Charts

Writing Process Analysis

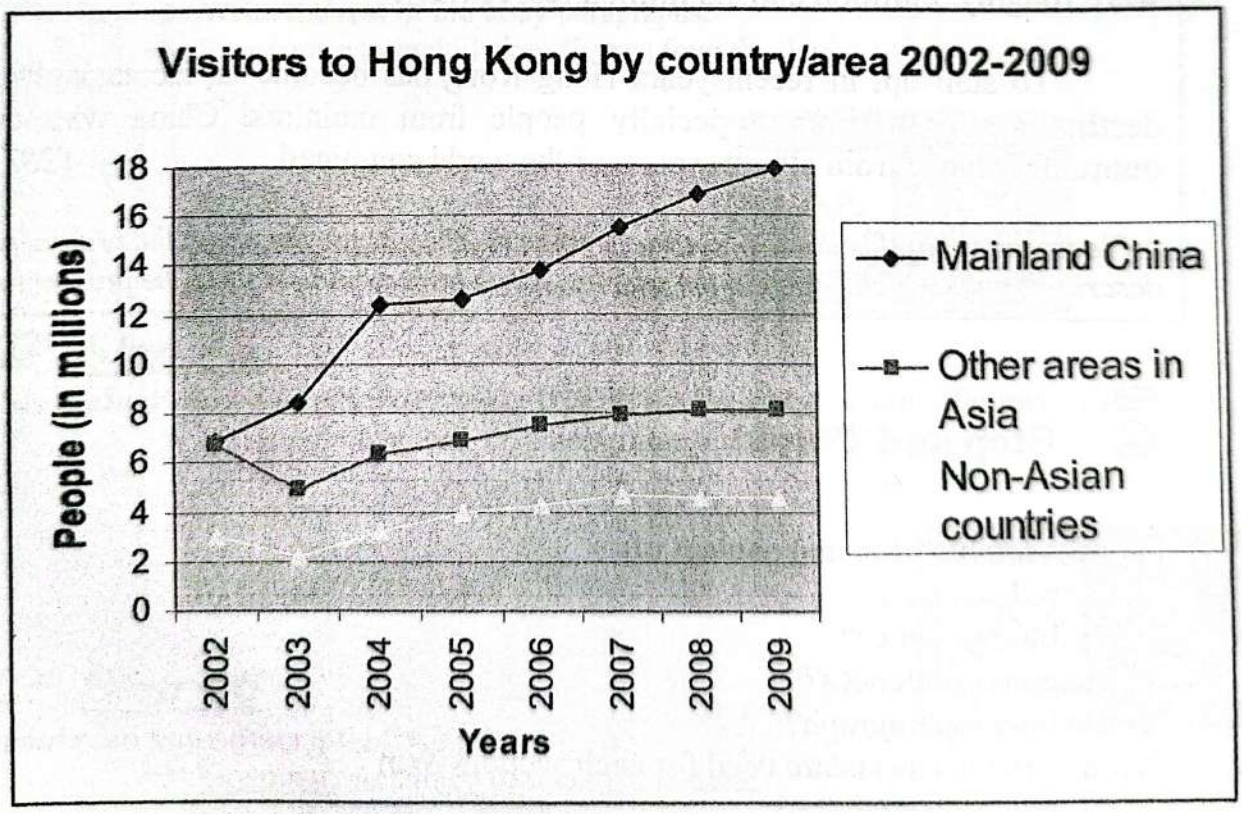
INTRODUCTION to writing a graph description



For the IELTS writing test, you are required to complete two writing tasks. Task 1 is some kind of data report, while Task 2 is an essay. You are probably already familiar with how to write an essay, but less familiar with the format and content of a report. So let's look at an example of a graph description, then break it down into steps so you can practice writing your own graph description reports for IELTS Task 1 practice.

NB: Not every Task 1 asks you to describe a graph; there may be tables, charts, or even a diagram or flowchart for you to describe instead. However, the same principles of organization and steps for writing the report apply.

FIRST SAMPLE IELTS TASK 1



Compiled from two sources (Last access date: August 3, 2009.):
http://www.censtatd.gov.hk/FileManager/EN/Content_807/transport.pdf
http://www.stats.gov.cn/tjsj/ndsj/2007/html/W2437E.HTM (Figures for 2009 have been extrapolated)

Sample Report 1 – Model Answer

The graph shows the number of travelers from various parts of the world entering Hong Kong for the years from 2002 to 2009. It can clearly be seen that there has been* a large increase in the number of Mainland Chinese visitors, while figures for visitors from Asia as well as other areas of the world have only shown* a slight growth.

The numbers of Mainland Chinese visiting Hong Kong have risen considerably over this time period. In 2002 there were nearly 7 million Mainland Chinese visitors, similar to the figure for other parts of Asia and just over double that for non-Asian travelers. This number climbed to approximately 8 million in 2003, then rose dramatically by 4 million to just over 12 million only one year later. Since 2004 there has been a steady increase, with around 18 million Mainland Chinese travelers visiting Hong Kong in 2009. This is more than double the number for that year of other visitors from Asia, and around four times the figures for non-Asian visitors.

The figures for travelers from other parts of the world show similar trends: first a dip of roughly one-third the total number of visitors from 2002 to 2003, then slow growth until 2007 followed by a leveling out. Overall, the numbers of Asian tourists rose by 1 million from just under 7 million in 2002 to roughly 8 million seven years later, while the corresponding figures for visitors from non-Asian parts of the world were roughly 3 million and 4.5 million respectively.

To sum up, in recent years Hong Kong has become an increasingly popular destination for visitors, especially people from mainland China who currently outnumber those from all other parts of the world combined. *(287 words)*

** Please note that this description was written in 2009, hence the use of the present perfect to describe trends which began in the past but which have continued up to the present time.*

Stop and Think!

What is the function and content of:
… the red sentence?
… the blue sentence?
… the green sentences?
… the brown paragraph?
What verb tense(s) is/are used for each section? Why?

Now let's read on and take a look at the steps that you could take to write a report like the above.

STEPS for writing a graph description

🕮 *Step #1*

Read the title and labels on the graph as well as the instructions.

→ Note down 2-3 synonyms for key words/phrases.

→ Write your introductory sentence, trying not to copy any phrases.

🕮 *Step #2*

Analyze the information in the graph, looking at the overall trend(s).

→ Write the rest of your introduction, explaining the main trend(s).

🕮 *Step #3*

Plan your body paragraphs (probably one paragraph for each main trend).

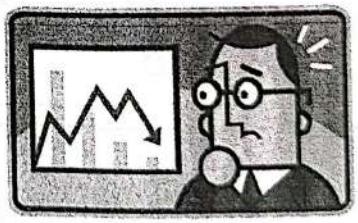
→ Draft a topic sentence for each body paragraph.

🕮 *Step #4*

Choose the most important details that will support your statement(s) of main trend(s).

→ Write the rest of the body paragraphs.

→ Remember to include key figures from the data.



🕮 *Step #5*

Think about how to restate the main trend(s) in the graph using different words.

→ Write your conclusion, paraphrasing the main trend(s).

🕮 *Step #6*

Proofread your description. Check:

→ <u>Grammar</u>: Accuracy of noun phrases and verb phrases, correct verb tenses & prepositions.

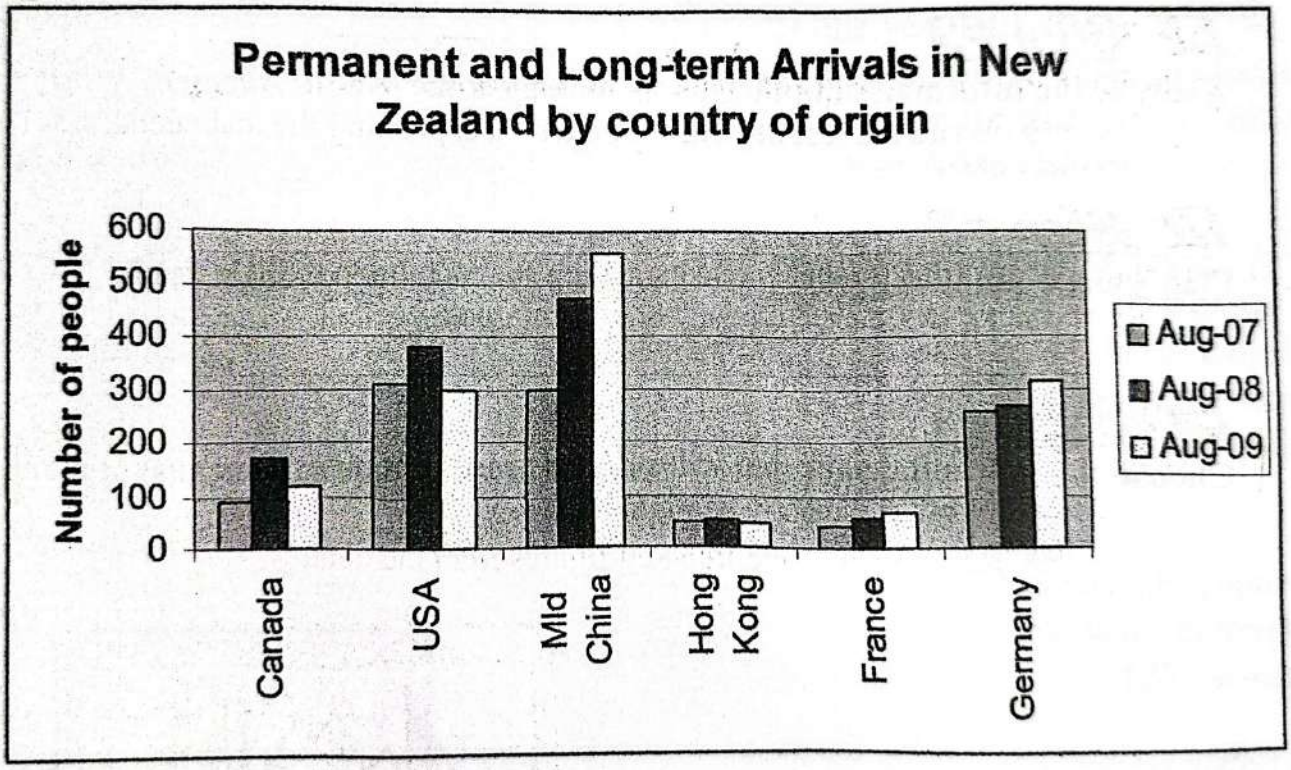
→ <u>Organization/coherence</u>: Linking words and variety of phrases/conjunctions.

→ <u>Content</u>: Is the information accurate and detailed enough? Do the details support the topic sentences?

Now try this process yourself, with a different graph. Work through the 6 steps. When you have finished, compare your answer with the sample answer (below).
(Note that this is not the only possible answer, nor the only possible way to describe the trends in this graph.)

SECOND SAMPLE IELTS TASK 1

The graph below shows permanent and long-term arrivals in New Zealand by country of origin in August of the years 2007 to 2009. Write a report of at least 150 words describing and comparing the data.



My notes

Sample Report 2 – Model Answer

The graph shows the number of immigrants to New Zealand from various parts of the world for the month of August in the years 2007 to 2009. Different trends can be seen for people arriving from North America, Asia and Europe, while immigrants from Mainland China showed the greatest increase as well as made up the largest proportion of immigrants throughout most of this time period.

The figures for people moving to New Zealand from different regions of the world showed diverse tendencies. The numbers of North American immigrants firstly peaked at a little over 150 for Canada and just under 400 for the USA in August 2008, then declined by around 60 to 80 the following year. Long-term arrivals in New Zealand from two European countries showed a continuous slight rise, with numbers of French and Germans reaching around 70 and 310 in 2009 respectively. In contrast, the figures for permanent arrivals from two Asian regions, Hong Kong and Mainland China, showed completely different trends. The number of Mainland Chinese immigrants grew sharply, almost doubling from about 300 in August 2007 to approximately 550 in 2009, comprising the largest group by 2009, around twice the size of the groups from America and Germany. On the other hand, numbers of Hong Kongers settling in New Zealand remained stable at roughly 50 per year, only around one-tenth the numbers of Mainland Chinese in 2009.

To sum up, areas in three different continents displayed different trends. The numbers for mainland Chinese immigrants rose most dramatically and in 2009 represented the largest group, while Hong Kongers comprised the smallest numbers.

(267 words)

Paragraphs to Describe Graphs and Charts

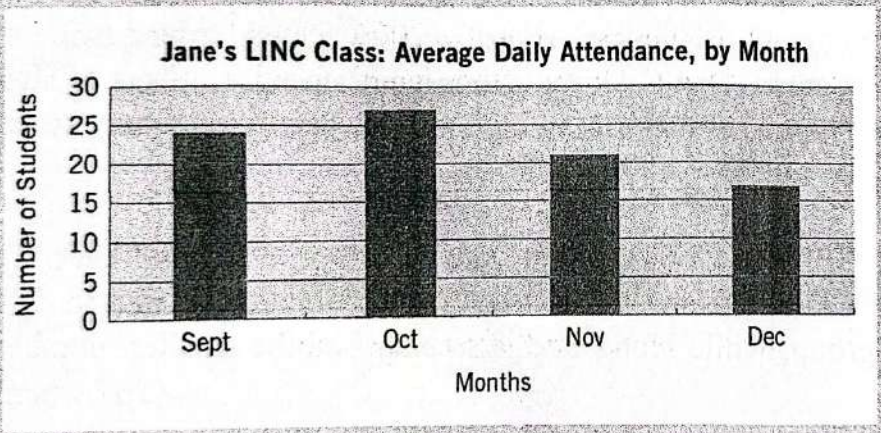
Often in academic writing, students will use information and facts presented in charts and graphs to support the topic sentence of their paragraph. Often, the data presented in a graph or table show change over time. There are specific verbs, adjectives and adverbs that can help you describe these changes. For example:

Verbs for describing how data have changed:

- *to increase*
- *to go up*
- *to rise*
- *to climb*
- *to remain (unchanged)*
- *to be steady*
- *to plateau*
- *to decrease*
- *to go down*
- *to decline*
- *to drop*

Adjectives and adverbs to describe how fast the change occurred:

- *sharp*
- *dramatic*
- *significant*
- *steady*
- *gradual*
- *slight*
- *sharply*
- *dramatically*
- *significantly*



You can describe the changes presented in the graph in two ways:

From October to December, attendance decreased steadily.

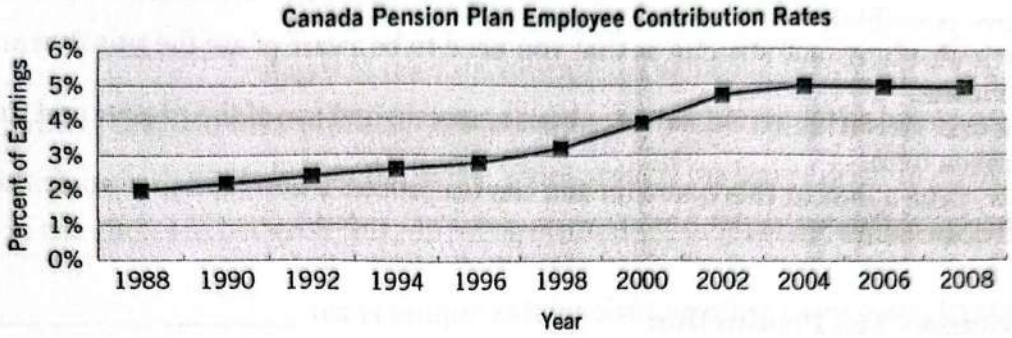
or

From October to December, there was a steady decrease in attendance.

1 **Write a paragraph to describe the bar graph above. Write three details to support the topic sentence.**

Topic sentence: Average attendance in Jane's LINC class followed a distinct pattern over time.

2 **Examine the graph below. Complete the paragraph that follows with appropriate supporting details.**



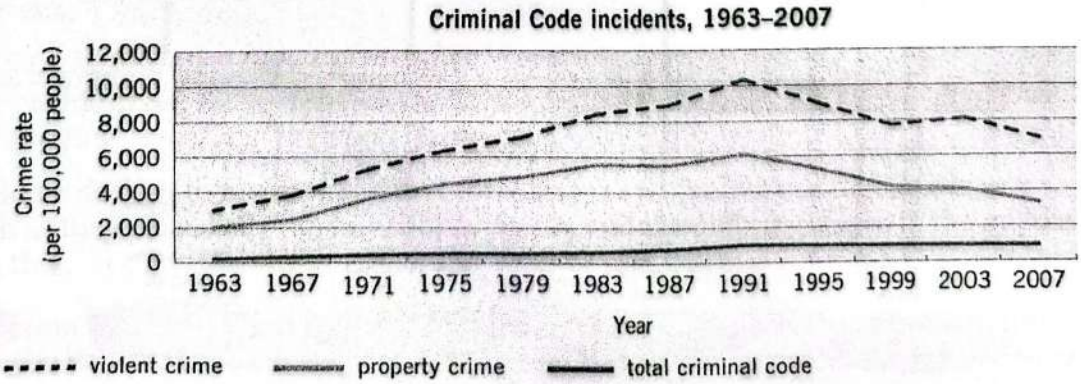
Data Source: Canada Revenue Agency

The Canada Pension Plan (CPP) is one of Canada's retirement income pension plans. It provides a monthly income to people (who contributed during their working years) when they become disabled or retire. Every person over 18 who earns more than about $3,500 a year contributes to the CPP, usually through paycheque deductions.

Employees contribute about 4.5 per cent of their employment earnings to the CPP, but the contribution rate was not always this high.

3 **Write a paragraph to describe the graph below. Begin with the provided topic sentence and prompts, and indicate possible reasons for the highs and lows in your supporting details.**

The crime rate has changed significantly over the past 45 years.



Data Source: Statistics Canada. Total criminal code refers to crimes addressed by criminal law and excludes rtraffic offences.

Process Analysis Writing

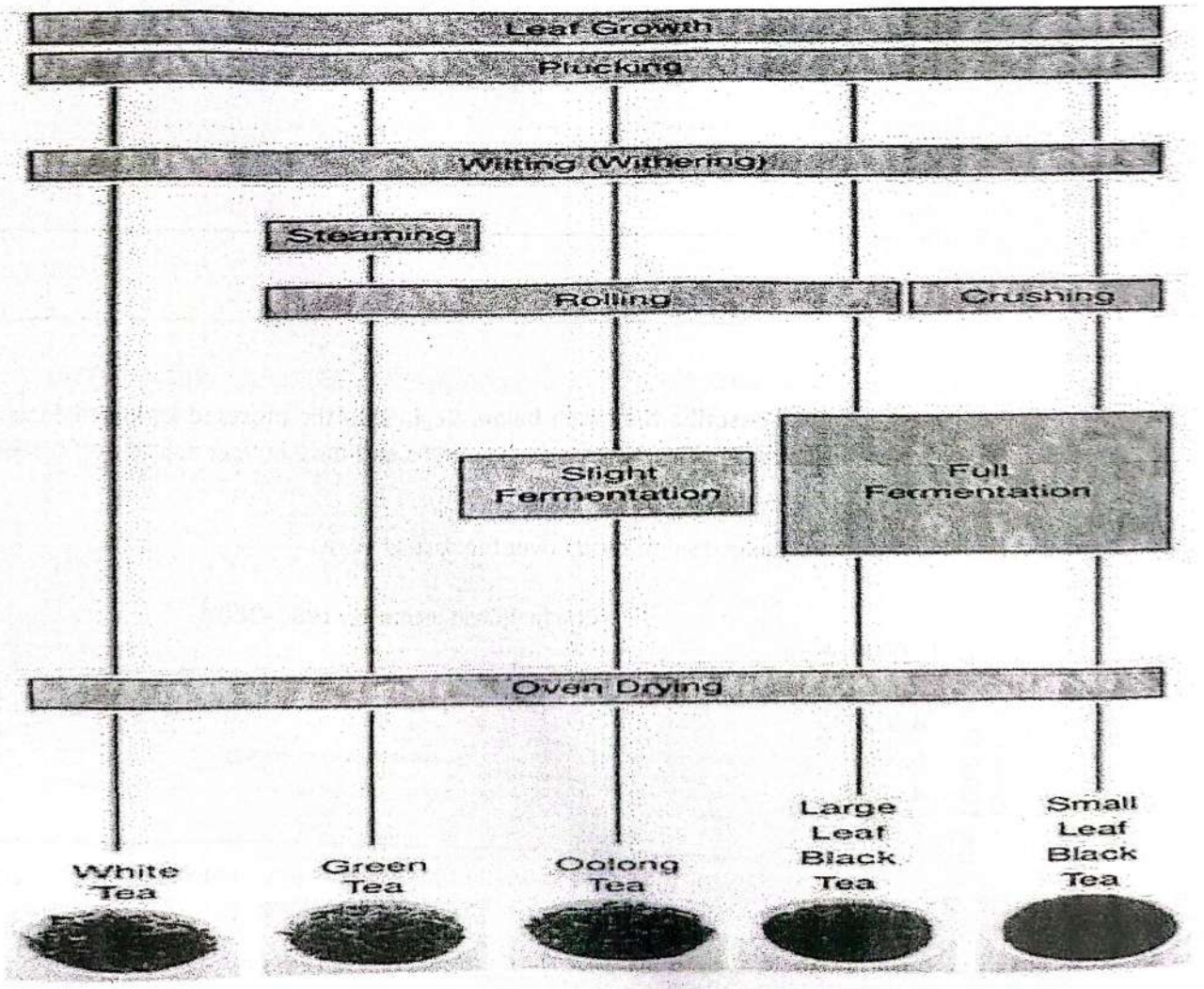
In this **process writing sample**, the manufacture of tea is described.
A process diagram is different to a graph or chart and requires some different language structures (though you still need to compare and contrast the various stages in the process where possible).
Particular language structures that you need to be aware of are the **passive** and **time phrases**.
You can view a lesson on writing about a process and use of the passive and time phrases here.
Now, take a look at this question and the tea process below, and then study the model answer and comments below.

Process - Tea Production

The diagram shows how tea leaves are processed into five tea types.

Summarize the information by selecting and reporting the main features and make comparisons where relevant.

Write at least 150 words.



Take clues from the diagram to complete the process description.

The diagram presents the manufacture of _____ different types of tea. It is immediately apparent that although all the types of tea are produced from the same leaf, the differences in the manufacturing process result in five different types of tea.

The first _____ stages of manufacture are the same for all of the five teas. The leaves are grown, they are then plucked, and following this withering of the leaves occurs. The _____ stage is also the same, which sees all the leaves dried in an oven. However, in the stages in between this, differing methods of production are employed.

To begin, _____ tea is unique as it involves no other processing. In contrast, green, oolong and large leaf black tea are all rolled as part of the process. However, while _____ tea is steamed before being rolled but is not fermented, the other two teas are first rolled and then both fermented (oolong only slightly but large leaf black completely). Finally, _____tea is neither steamed nor rolled, but is crushed before being fully fermented.

Comments

This process writing sample is a well-organized response that covers all the important features of the diagram.

It is easy to follow as it discusses each tea in turn, but not only this, it also groups similar processes together and identifies the differences.

For example:

Similarities:

The first three stages of manufacture are the same for all of the five teas.

...green, oolong and large leaf black tea are all rolled as part of the process.

Differences:

...white tea is unique as it involves no other processing.

However, while green tea is steamed before being rolled but is not fermented, the other two teas are first rolled and then both fermented.

The candidate decides to mention the first three stages and the last one together in one paragraph as they are exactly the same for every tea, and the differing stages in the middle following this.

This may seem odd to mention the stages in this order, and it is likely that when you describe a process you will discuss each stage in turn, but for this particular process it works well as the candidate can then focus on the differences.

The appropriate voice is also used in the description, which is the **passive.**

When we write about a process, we are interested in the actions, NOT who is doing them. In this case we use the passive voice. Here are some examples of the passive voice from this writing sample process:

...the teas are produced from the same leaf
The leaves are grown, they are then plucked
...differing methods of production are employed

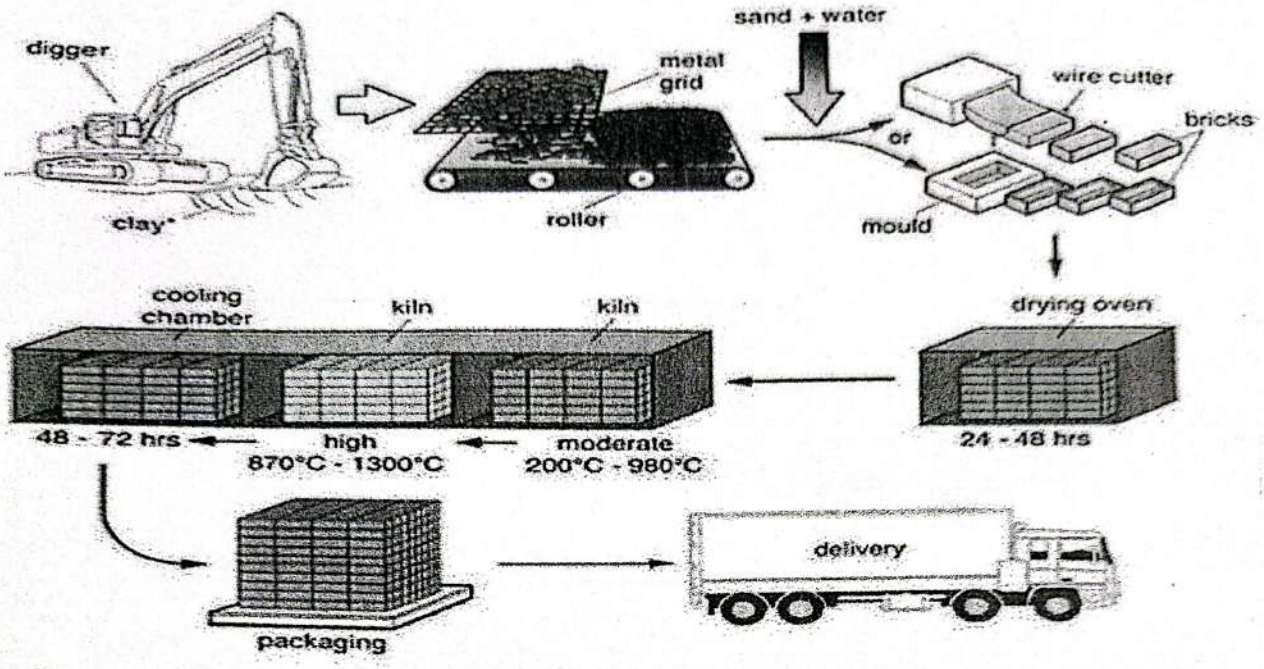
It may not always be possible, but if you can you should also try to use **synonyms** for some of the words from the diagram rather than copying them all.

For example, '*completely*' fermented is used instead of '*fully*'.

Make sure you also make use of a mix of **'time' transitions** to guide the reader through the description:

*...they are then plucked, and **following this** withering of the leaves occurs.*
*...the other two teas are **first** rolled and **then** both fermented*

The following diagram describes how bricks are produced for building industry. Fill in the gaps with the most appropriate words to complete the description.



The given diagram shows different stages of a)_____ for the building industries. It is apparent that brick production involves b) _____ steps, starting from the digging stage and ending at the delivery stage.

As is given in the graph, the first step of brick manufacturing is digging the clay with a large digger. This clay is then filtered and processed in a roller machine and then mixed with c)_____. In the third stage, the mixture is either kept in a mould or cut in a wire cutter to make the raw shape of the bricks. In the next step, the shaped raw bricks are dried for 24 to 48 hours in a d) _____.

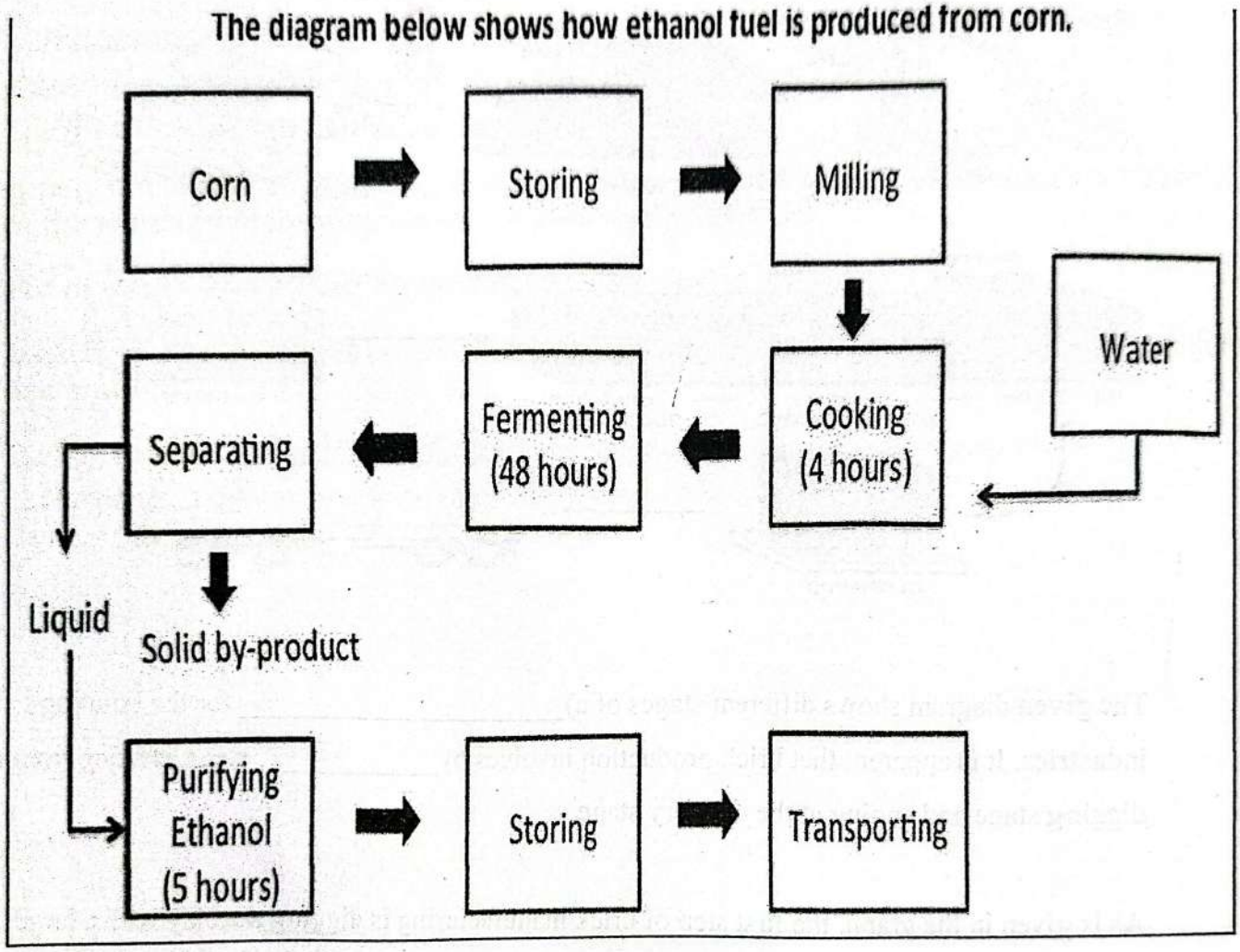
After that, the dried raw bricks are e) _____ in a kiln, both in moderate and high temperature, up to 1300 degree centigrade and then placed in a cooling chamber for about f) _____. This process makes the brick to be g) _____ in the final two steps.

The delivery process is the final process of brick manufacturing and as it is noted, brick making is a moderately complex process that requires some predefined works to make it usable.

Process Writing Practice

The diagram below shows how ethanol fuel is produced from corn.

Write at least 150 words.



Suggestions:

[Use passive voice and time transitions such as first, next, after that, then, after, before, finally]

– Introduction: (1) Introduce the topic, (2) the number of stages (3) refer to the first and last stages

– Paragraph 2: report the stages from the storage of corn until fermentation

– Paragraph 3: report the stages after fermentation until the end of the process